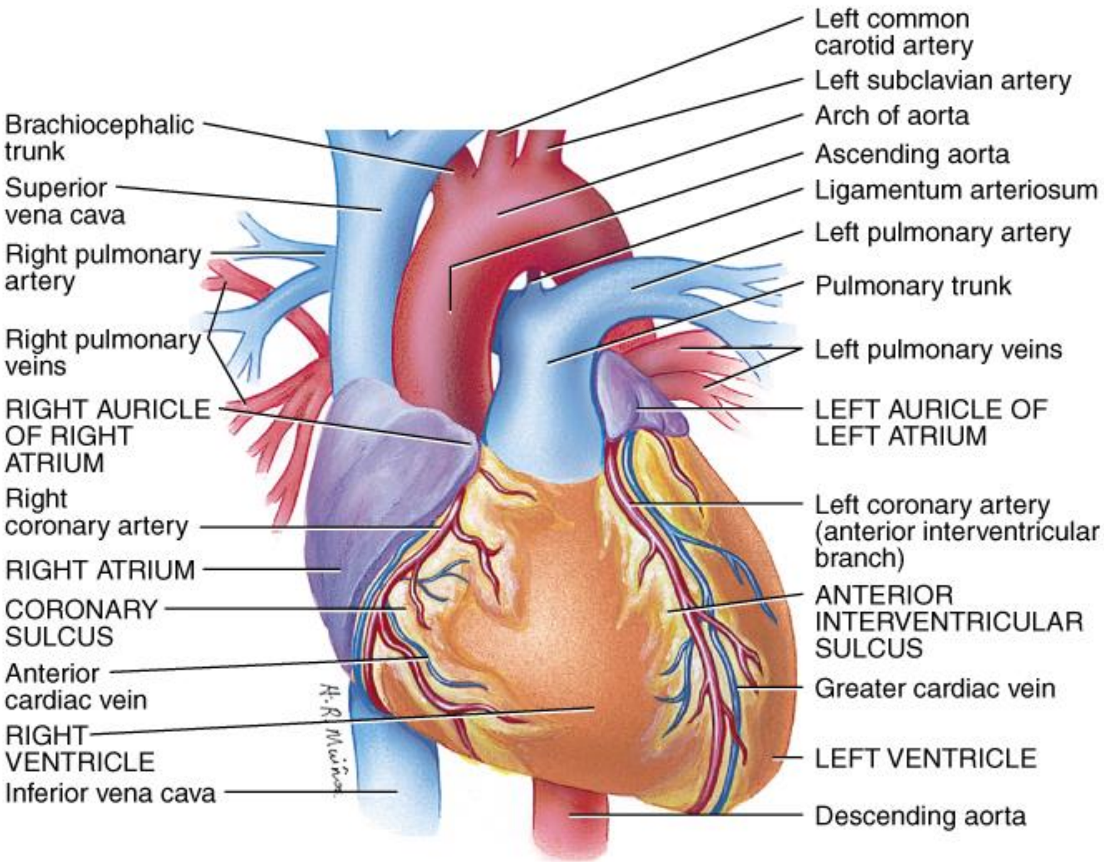
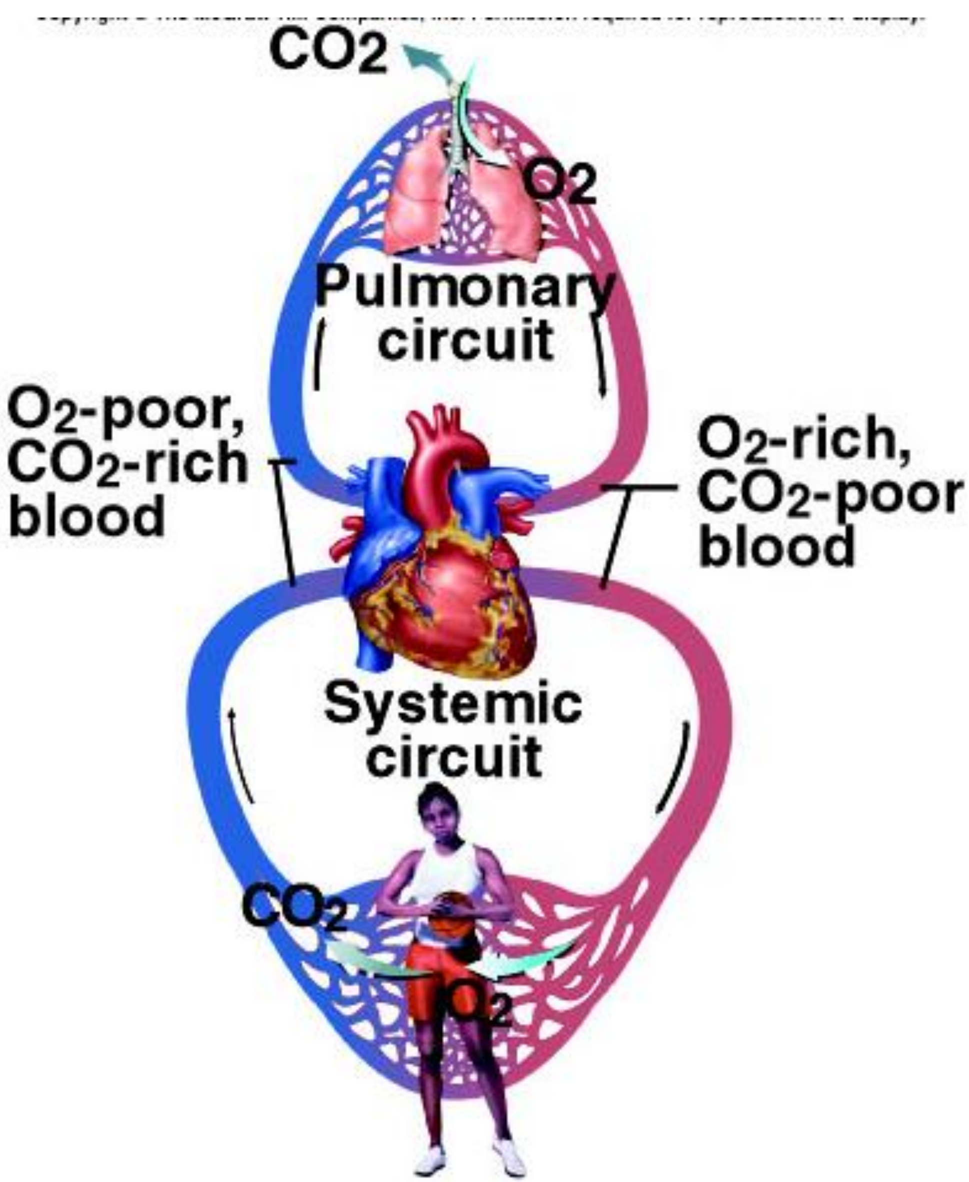
The Cardiovascular System: The Heart



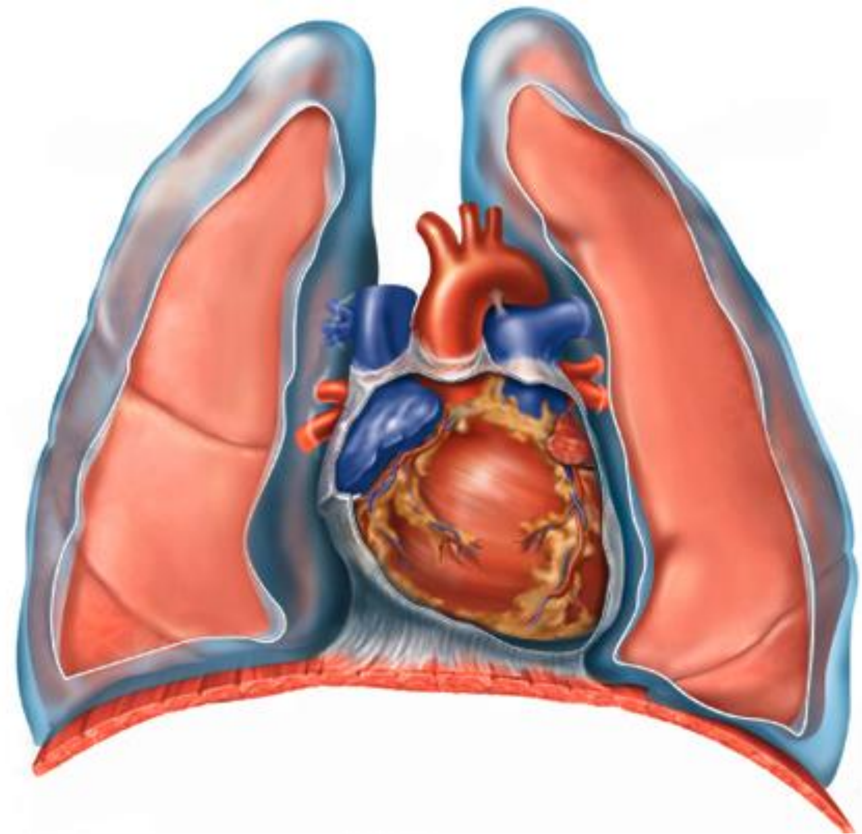
- Heart pumps over 1 million gallons per year
- Over 60,000 miles of blood vessels

Cardiovascular System Circuit

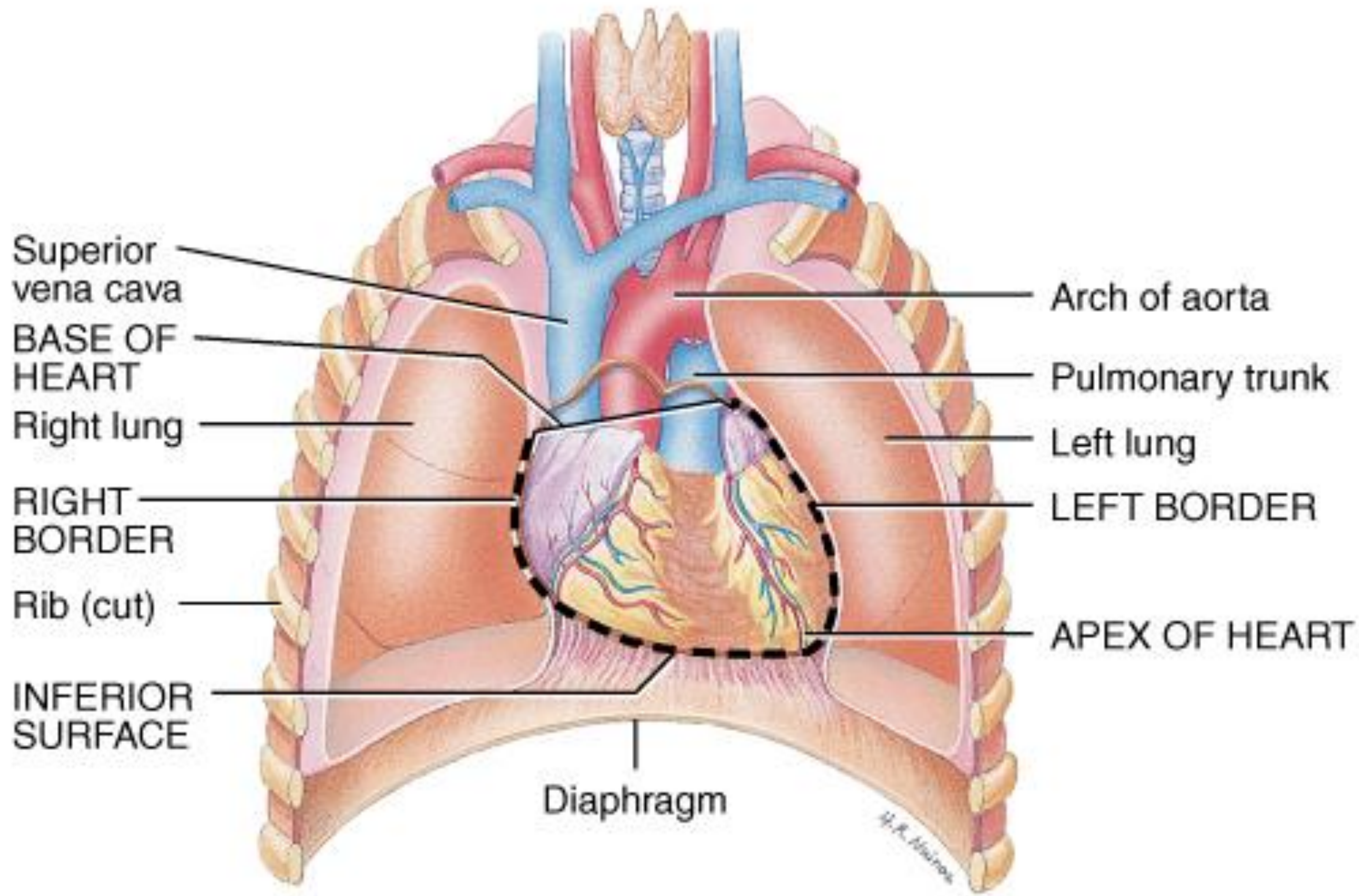


Size, Shape and Position

- Located in mediastinum, between lungs
- Base - broad superior portion of heart
- Apex - inferior end, tilts to the left, tapers to point
- 3.5 in. wide at base, 5 in. from base to apex and 2.5 in. anterior to posterior; weighs 10 oz

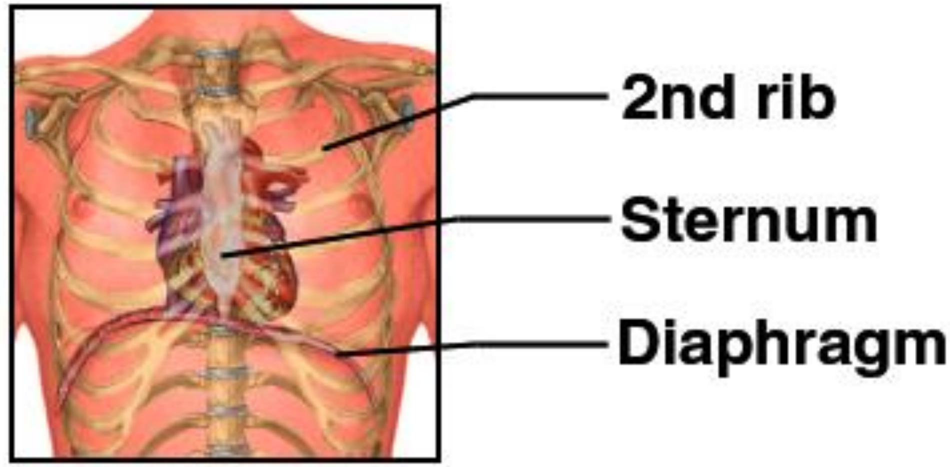


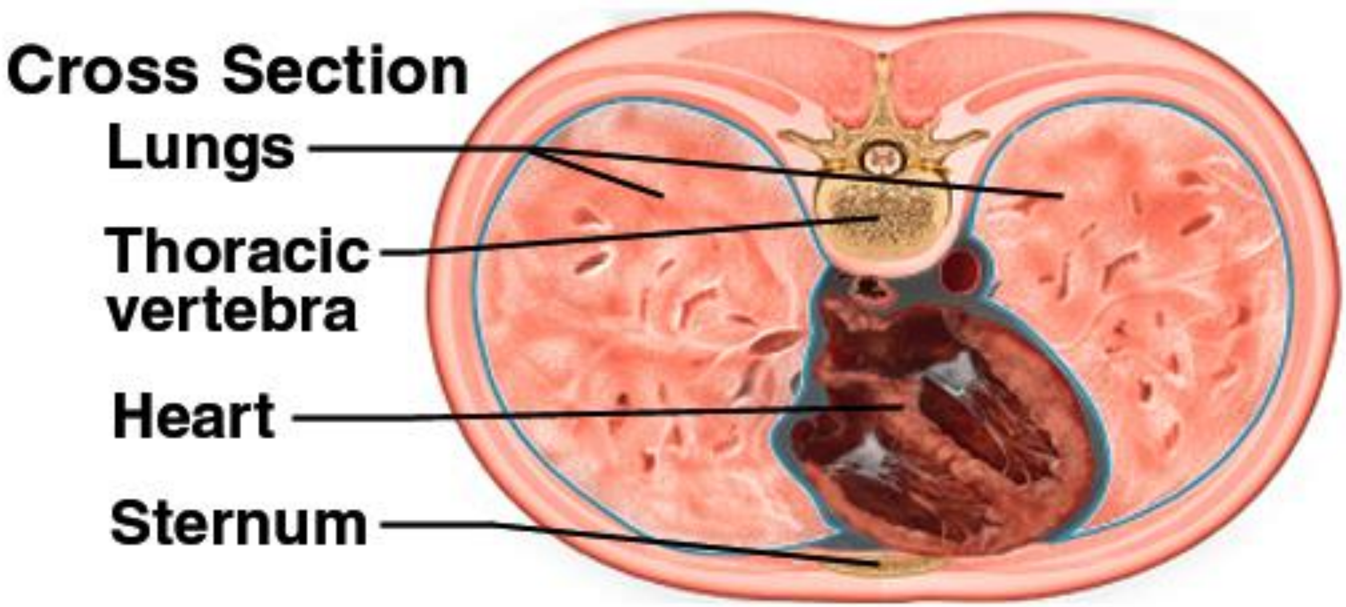
Heart Orientation



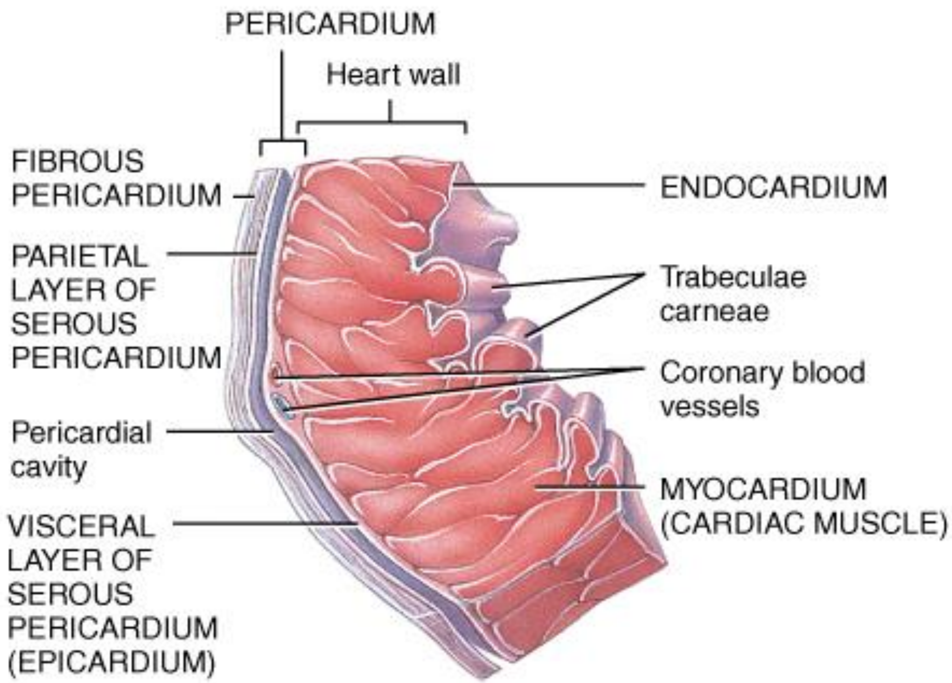
- Heart has 2 surfaces: anterior and inferior, and 2 borders: right and left

Heart Orientation





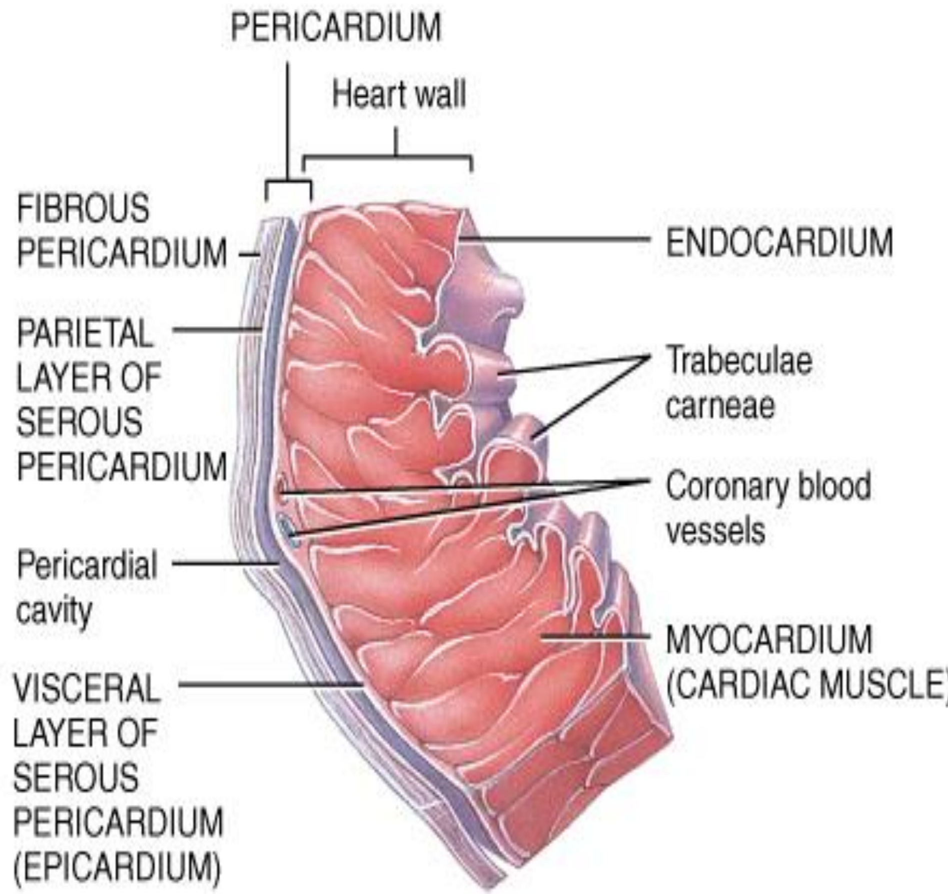
Pericardium



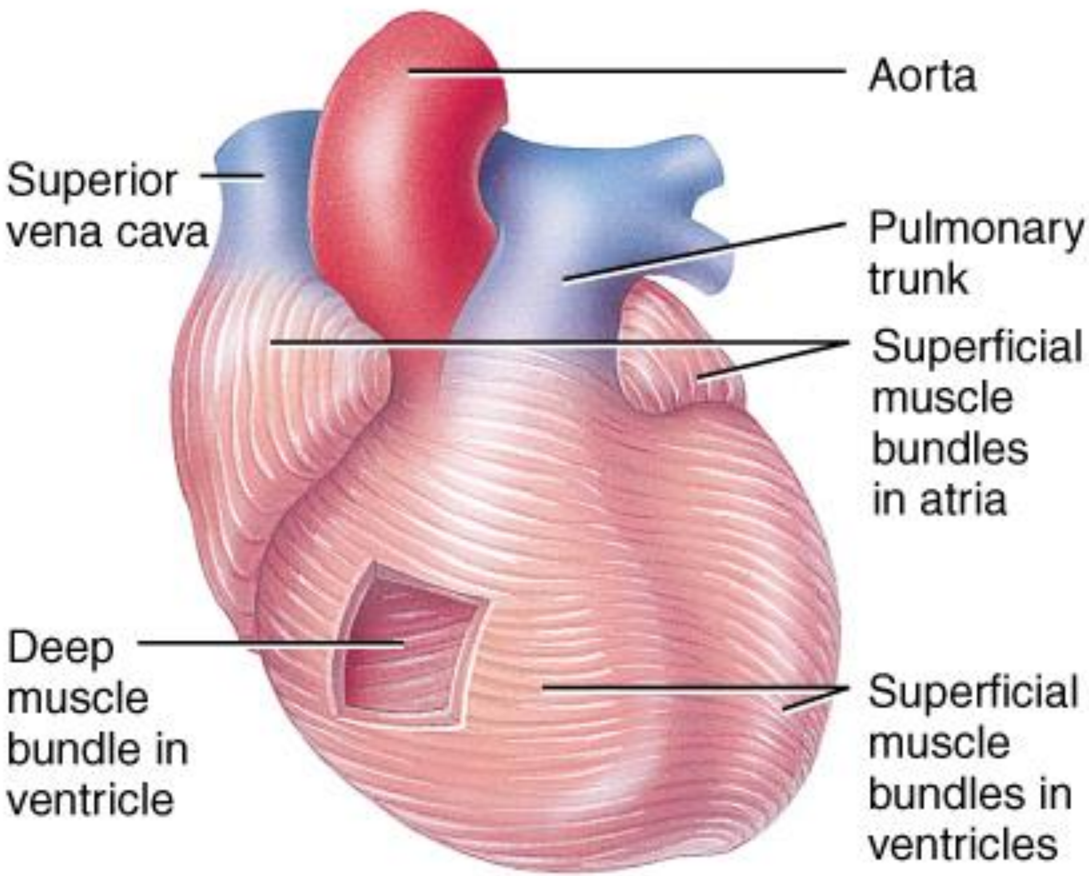
- Fibrous (parietal) pericardium
  - dense irregular CT
  - protects and anchors the heart, prevents overstretching
- Serous (visceral) pericardium
  - thin delicate membrane
  - contains
    - parietal layer-outer layer
    - pericardial cavity with pericardial fluid
    - visceral layer (epicardium)

Heart Wall

- Epicardium (a.k.a. visceral pericardium)
  – serous membrane covers heart
- Myocardium
  – thick muscular layer
  – fibrous skeleton - network of collagenous and elastic fibers
    - provides structural support
    - attachment for cardiac muscle
    - nonconductor important in coordinating contractile activity
- Endocardium
  – smooth inner lining



Muscle Bundles of the Myocardium

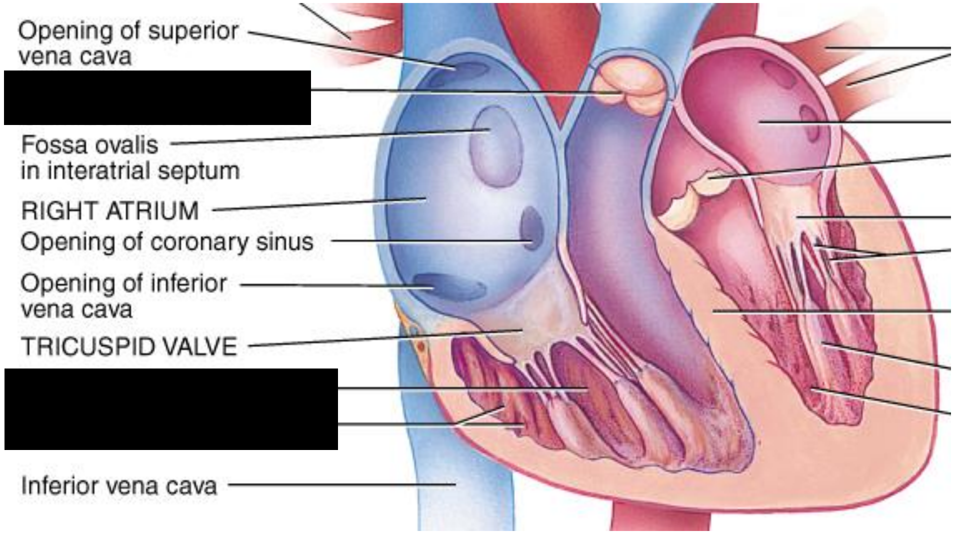


- Cardiac muscle fibers swirl diagonally around the heart in interlacing bundles

Chambers and Sulci of the Heart

- Four chambers
  - 2 superior and posterior atria receive returning blood
  - 2 inferior and anterior ventricles pump outgoing blood
- Sulci - grooves on surface of heart containing coronary blood vessels and fat
  - coronary sulcus
    - encircles heart and marks the boundary between the atria and the ventricles
  - anterior interventricular sulcus
    - marks the boundary between the ventricles anteriorly
  - posterior interventricular sulcus
    - marks the boundary between the ventricles posteriorly

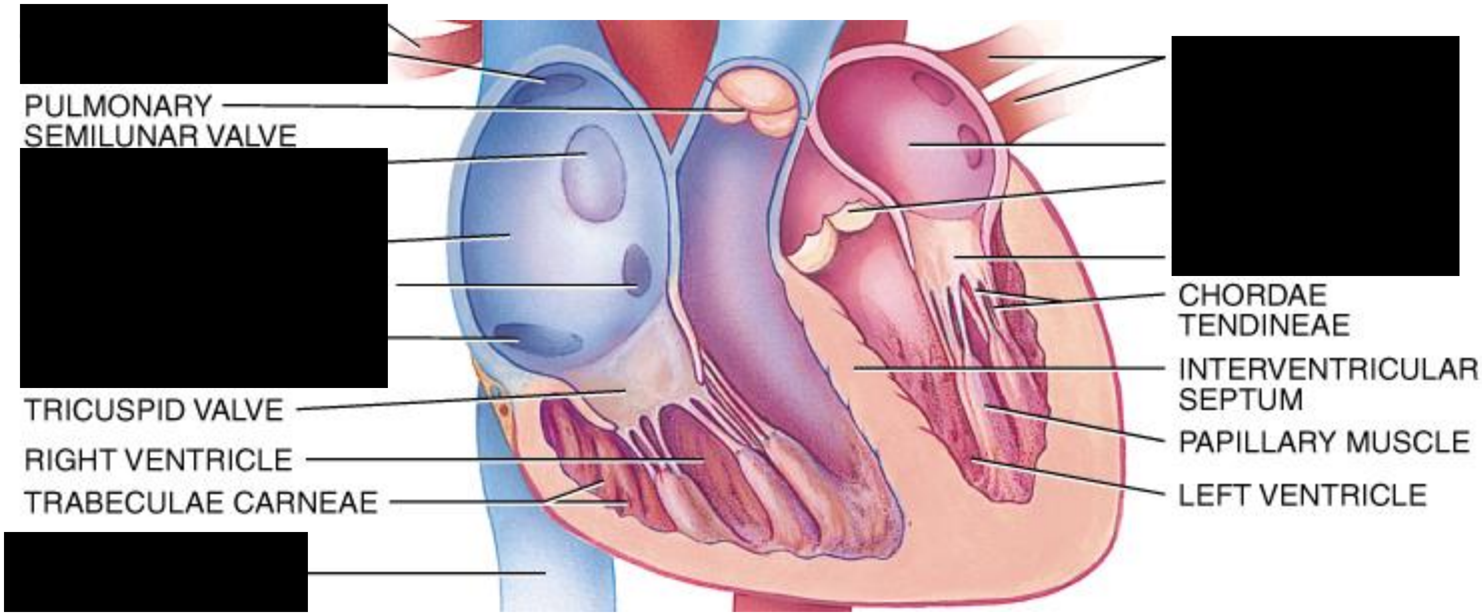
Right Atrium



- Receives blood from 3 sources
  - superior vena cava, inferior vena cava and coronary sinus
- Interatrial septum partitions the atria
- Fossa ovalis is a remnant of the fetal foramen ovale
- Tricuspid valve
  - Blood flows through into right ventricle
  - has three cusps composed of dense CT covered by endocardium

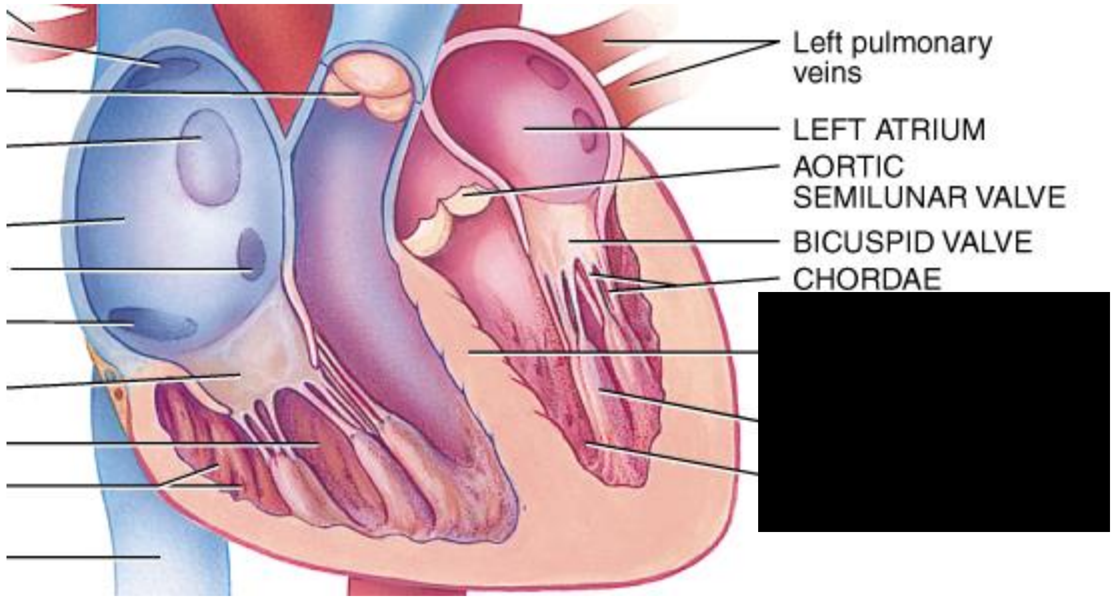
Right Ventricle

[redacted]



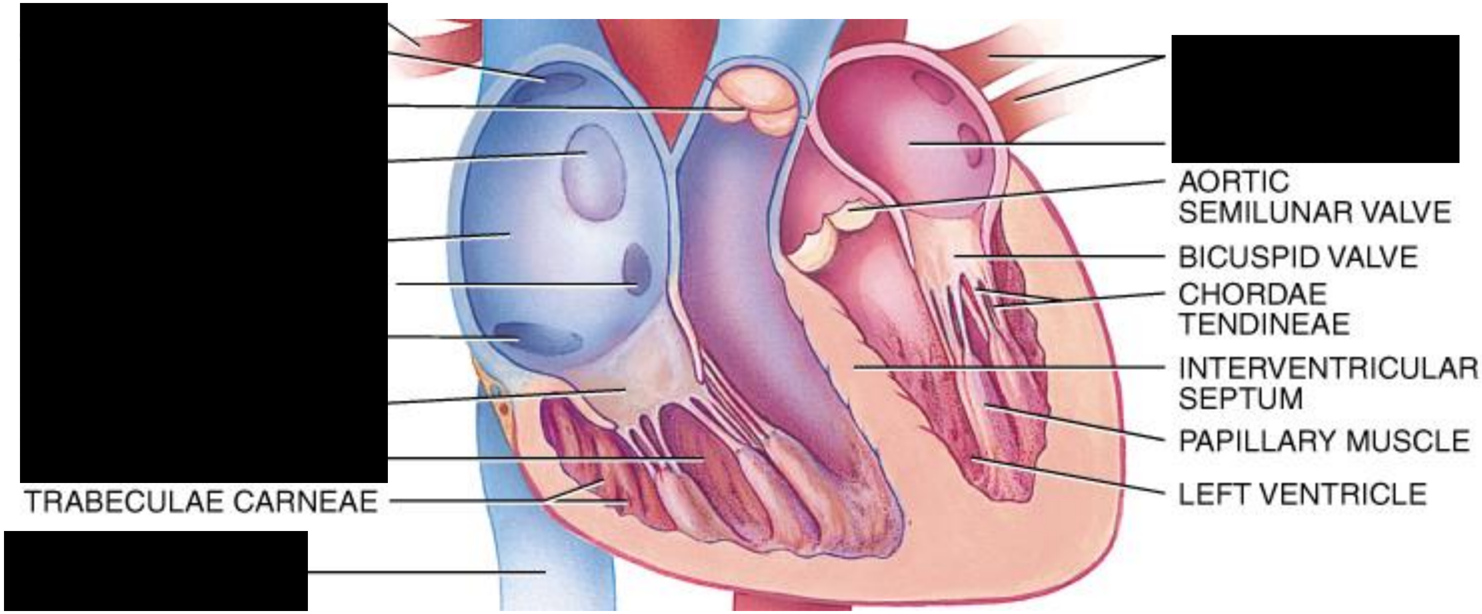
- Forms most of anterior surface of heart
- Papillary muscles are cone shaped trabeculae carneae (raised bundles of cardiac muscle)
- Chordae tendineae: cords between valve cusps and papillary muscles
- Interventricular septum: partitions ventricles
- Pulmonary semilunar valve: blood flows into pulmonary trunk

Left Atrium



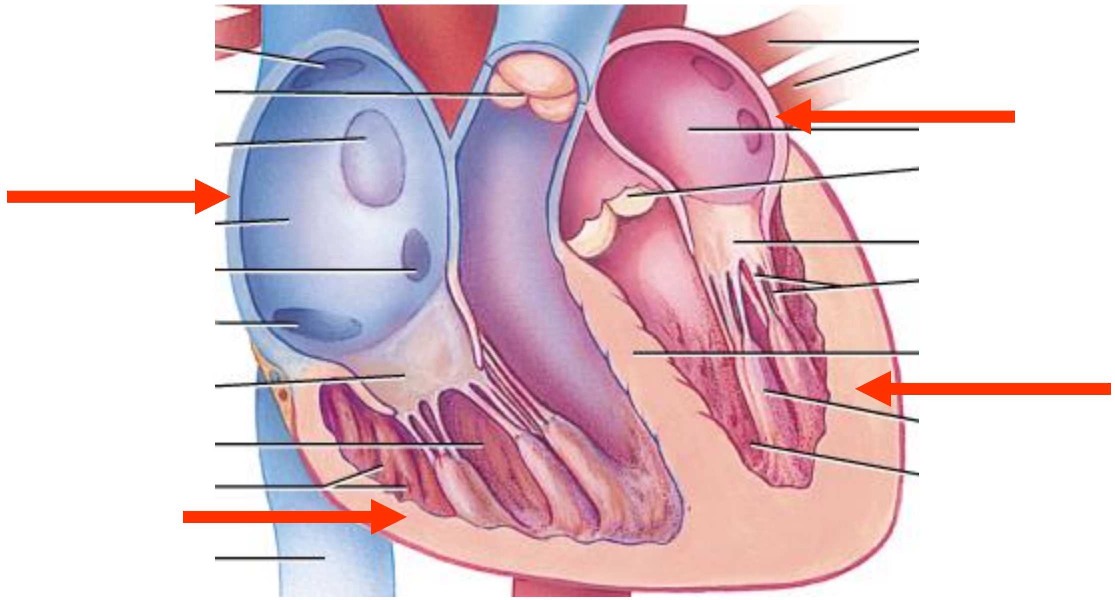
- Forms most of the base of the heart
- Receives blood from lungs - 4 pulmonary veins (2 right + 2 left)
- Bicuspid valve: blood passes through into left ventricle
  – has two cusps
  – to remember names of this valve, try the pneumonic LAMB
    - Left Atrioventricular, Mitral, or Bicuspid valve

Left Ventricle



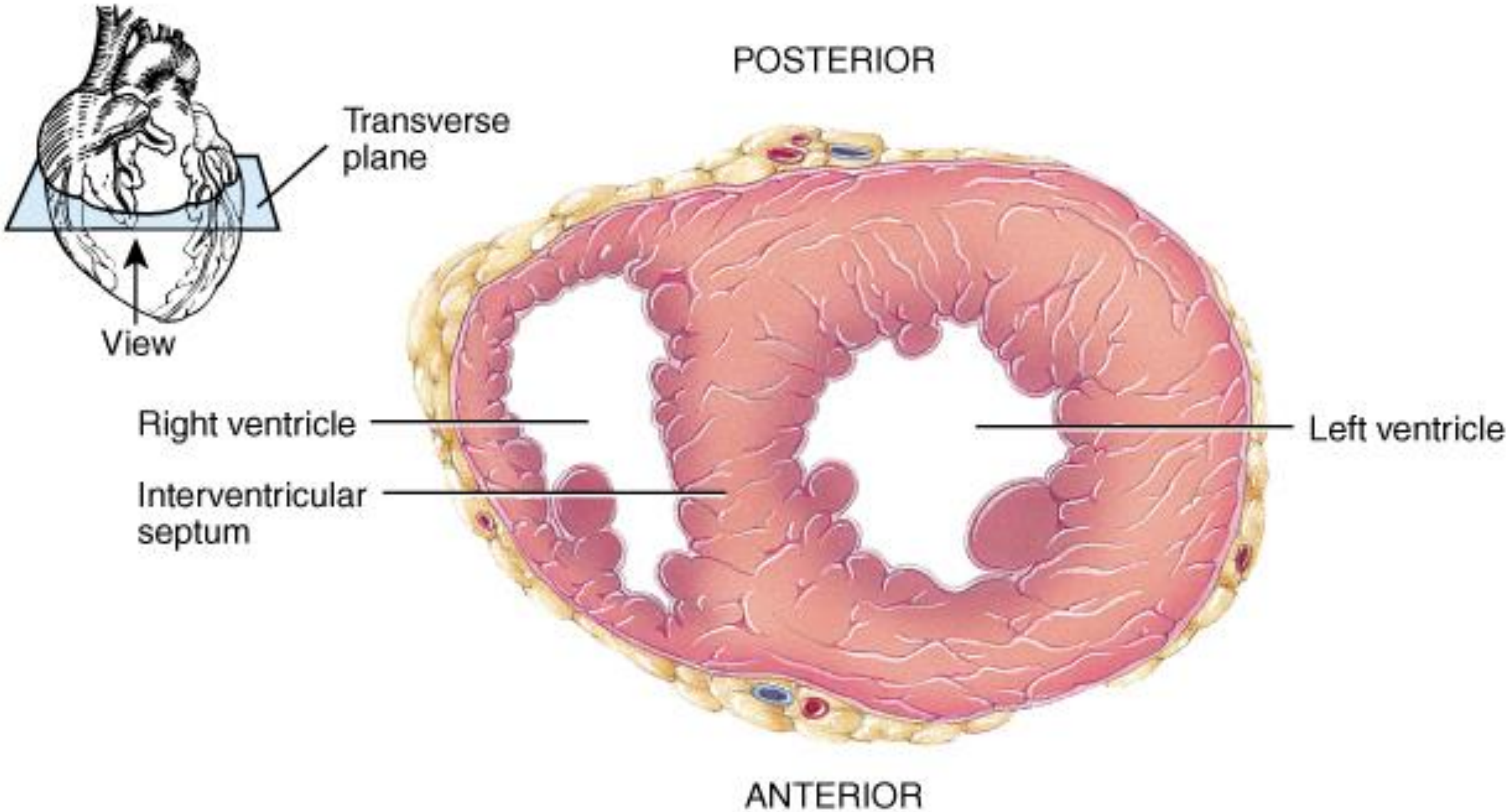
- Forms the apex of heart
- Chordae tendineae anchor bicuspid valve to papillary muscles (also has trabeculae carneae like right ventricle)
- Aortic semilunar valve:
  - blood passes through valve into the ascending aorta
  - just above valve are the openings to the coronary arteries

Myocardial Thickness and Function



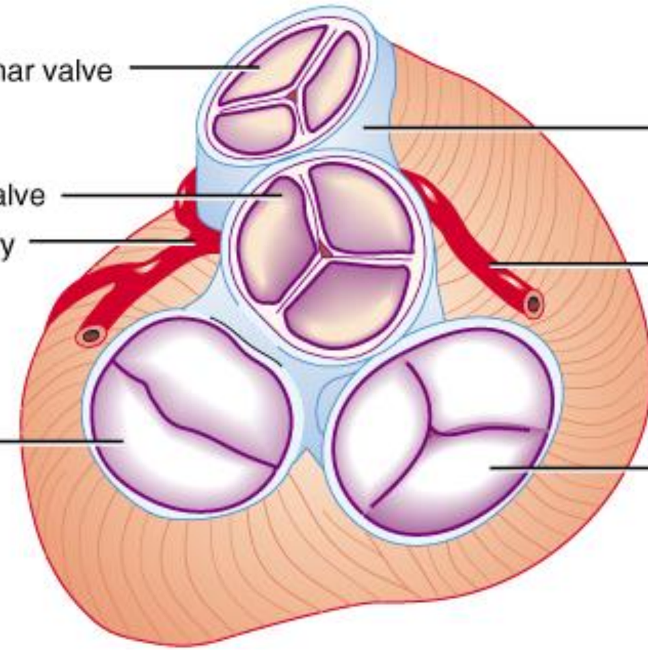
- Thickness of myocardium varies according to the function of the chamber
- Atria are thin walled, deliver blood to adjacent ventricles
- Ventricle walls are much thicker and stronger
  - right ventricle supplies blood to the lungs (little flow resistance)
  - left ventricle wall is the thickest to supply systemic circulation

Thickness of Cardiac Walls



Myocardium of left ventricle is much thicker than the right.

Fibrous Skeleton of Heart

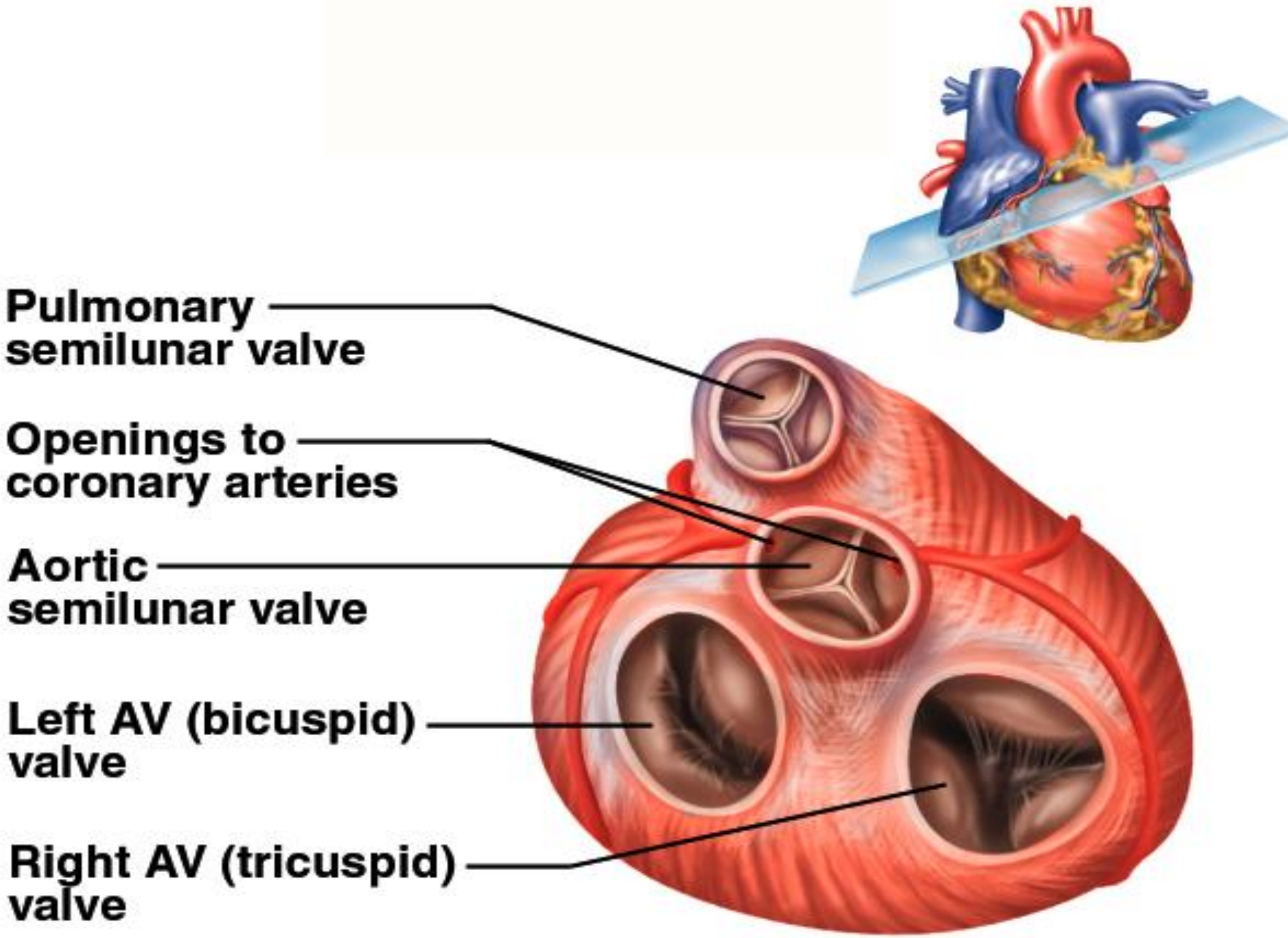


- Dense CT rings surround the valves of the heart, fuse and merge with the interventricular septum
- Support structure for heart valves
- Insertion point for cardiac muscle bundles
- Electrical insulator between atria and ventricles
  - prevents direct propagation of AP's to ventricles

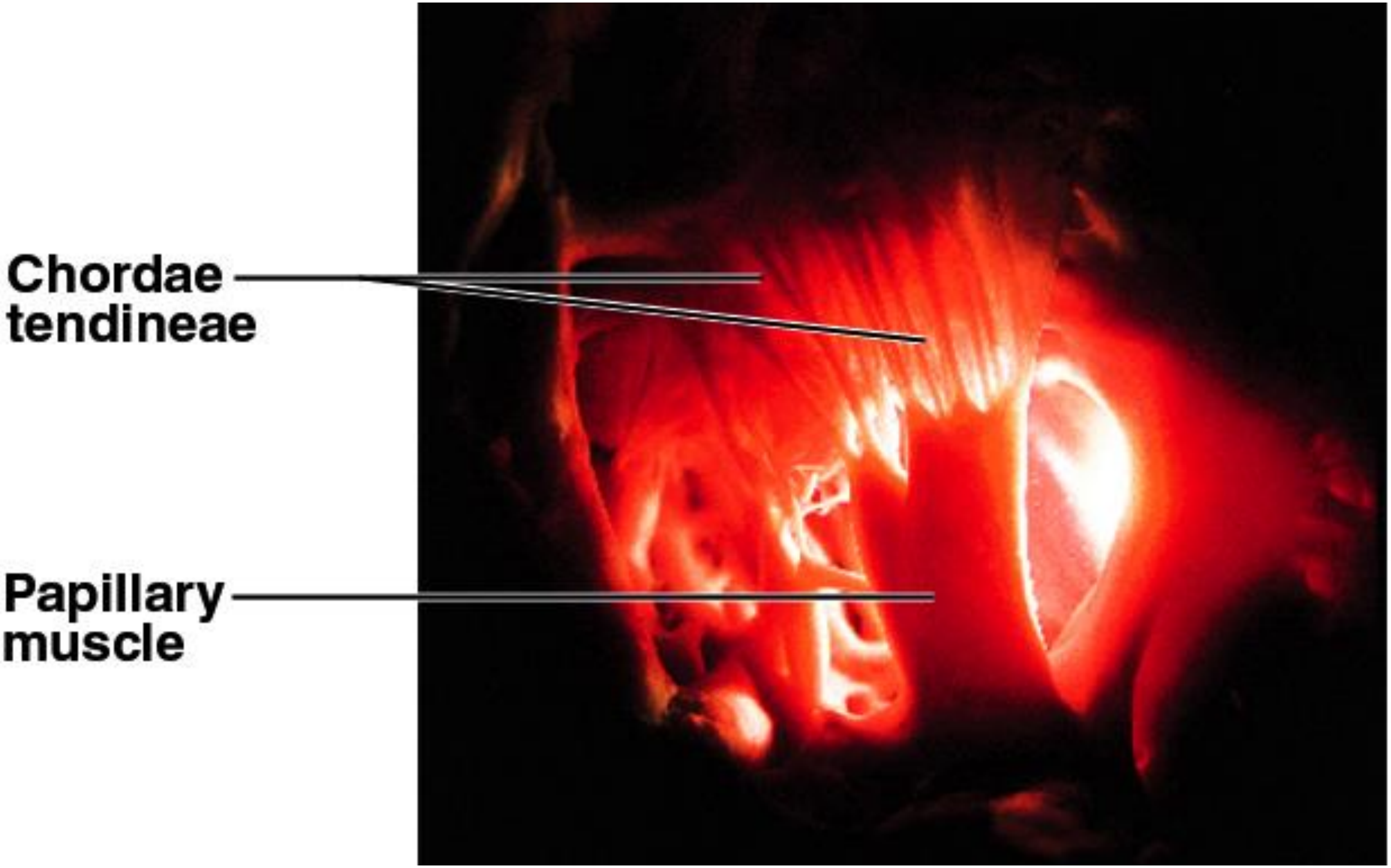
Heart Valves

- Ensure one-way blood flow
- Atrioventricular (AV) valves
  - right AV valve has 3 cusps (tricuspid valve)
  - left AV valve has 2 cusps (mitral, bicuspid valve) lamb
  - chordae tendineae - cords connect AV valves to papillary muscles (on floor of ventricles)
- Semilunar valves - control flow into great arteries
  - pulmonary: from right ventricle into pulmonary trunk
  - aortic: from left ventricle into aorta

Heart Valves



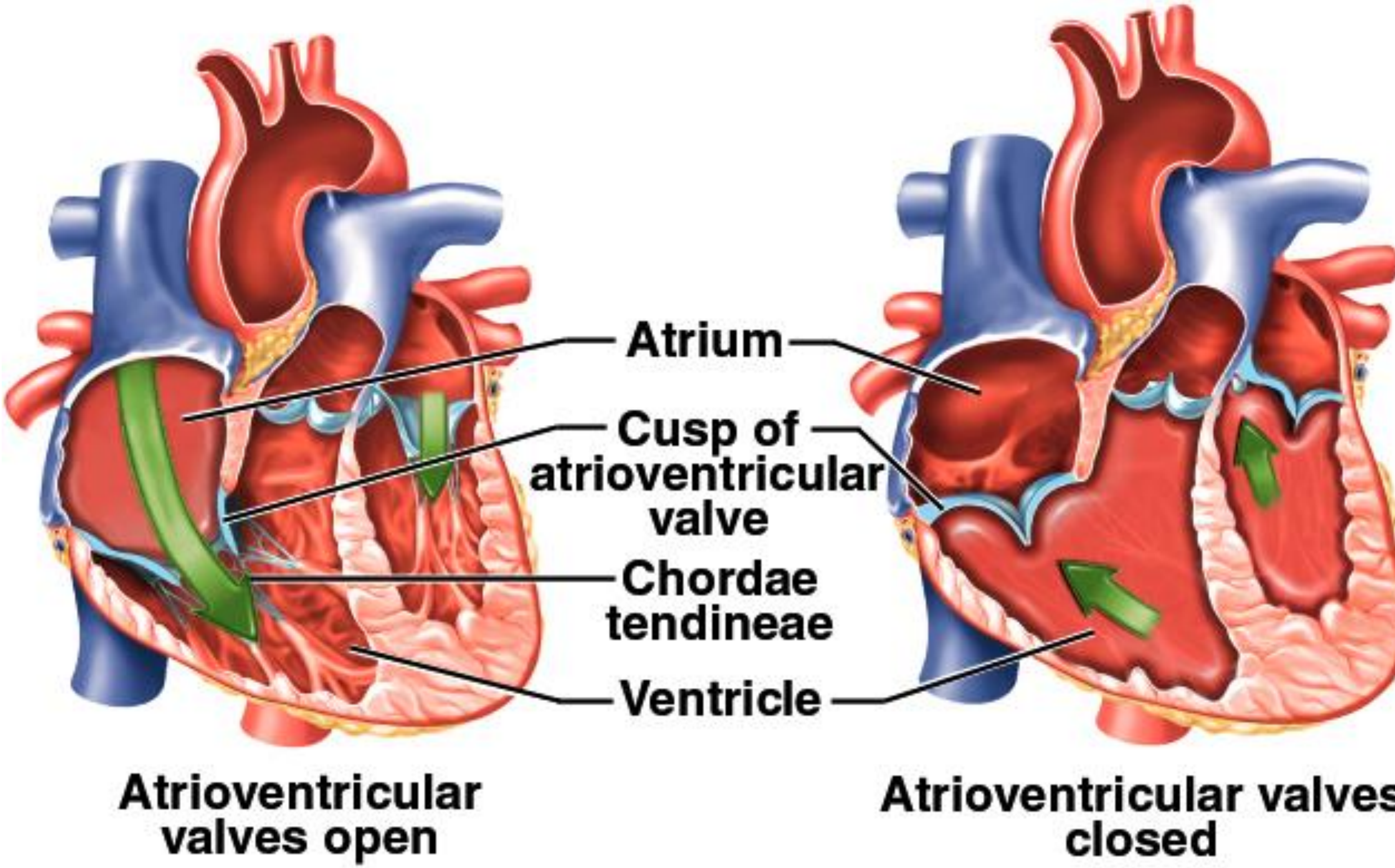
Heart Valves



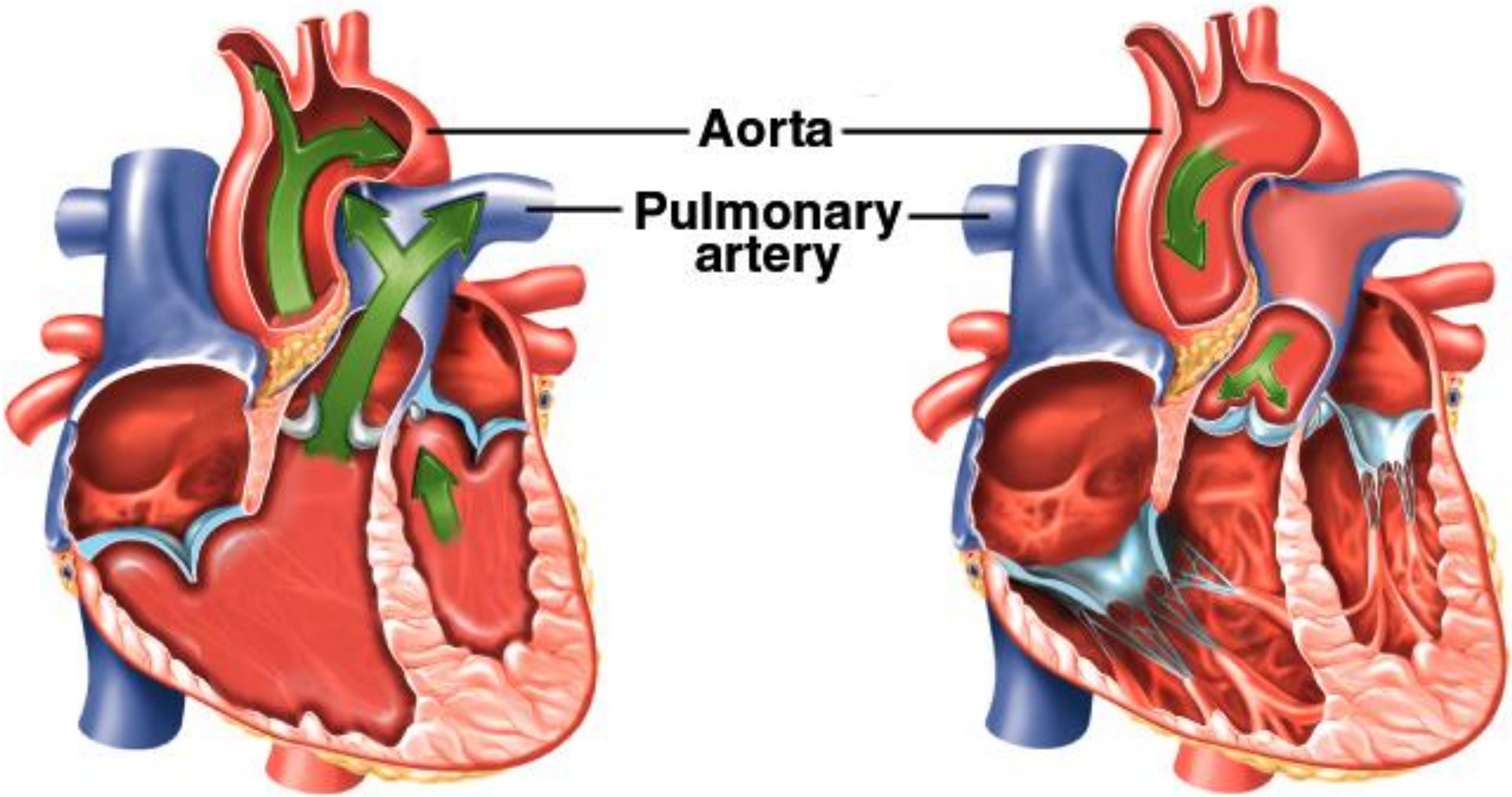
AV Valve Mechanics

- Ventricles relax, *pressure drops*, semilunar valves close, AV valves open, blood flows from atria to ventricles
- Ventricles contract, AV valves close (papillary m. contract and pull on chordae tendineae to prevent prolapse), *pressure rises,* semilunar valves open, blood flows into great vessels

Operation of Atrioventricular Valves



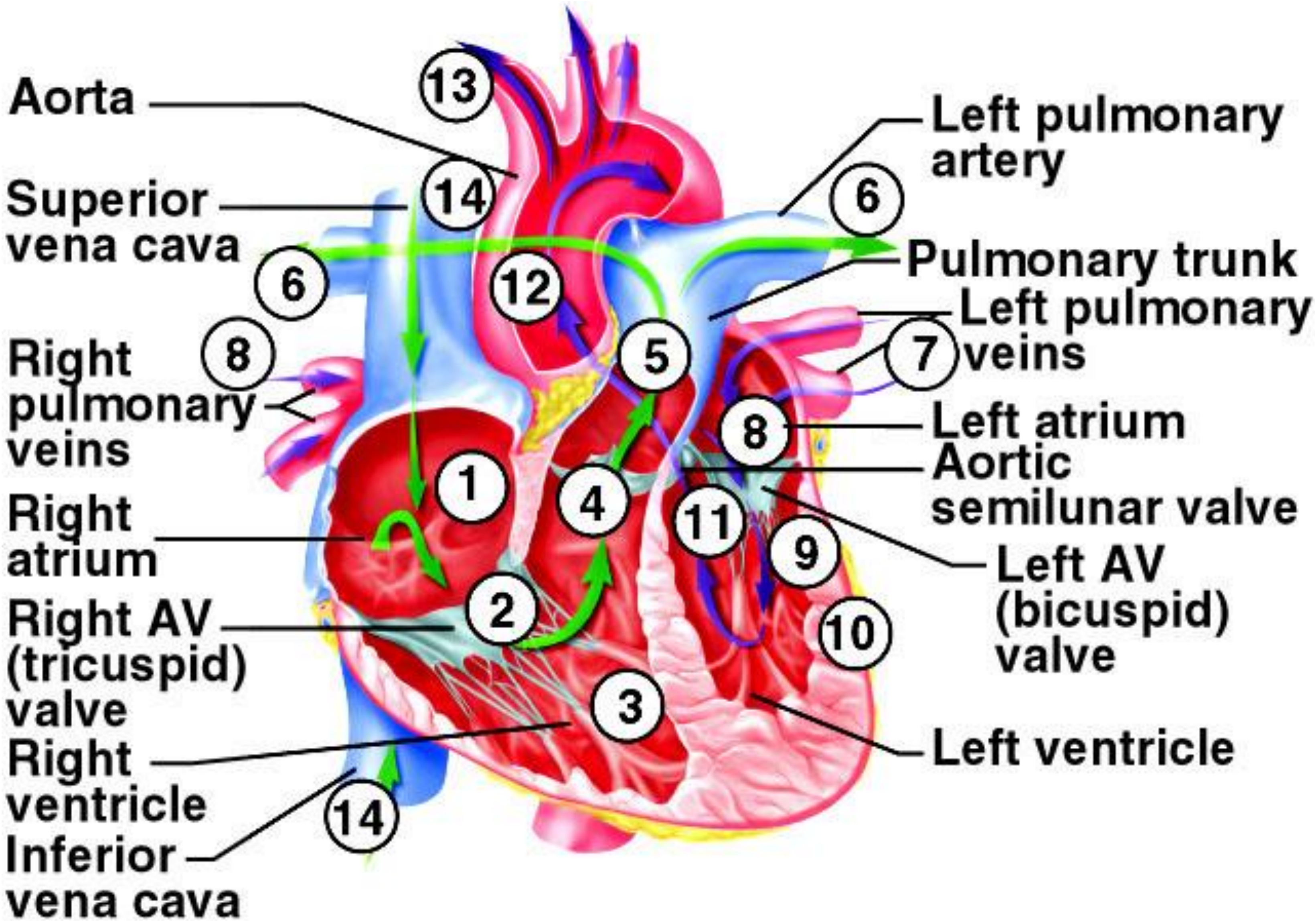
Operation of Semilunar Valves



Semilunar valves open

Semilunar valves closed

Blood Flow Through Heart



Coronary Circulation

- Coronary circulation is blood supply to the heart
- Heart as a very active muscle needs lots of O_2
- When the heart relaxes high pressure of blood in aorta pushes blood into coronary vessels
- Many anastomoses
  - connections between arteries supplying blood to the same region, provide alternate routes if one artery becomes occluded

Myocardial Infarction

- Sudden death of heart tissue caused by interruption of blood flow from vessel narrowing or occlusion
- Anastomoses defend against interruption by providing alternate blood pathways
  - circumflex artery and right coronary artery combine to form posterior interventricular artery
  - anterior and posterior interventricular arteries combine at apex of heart

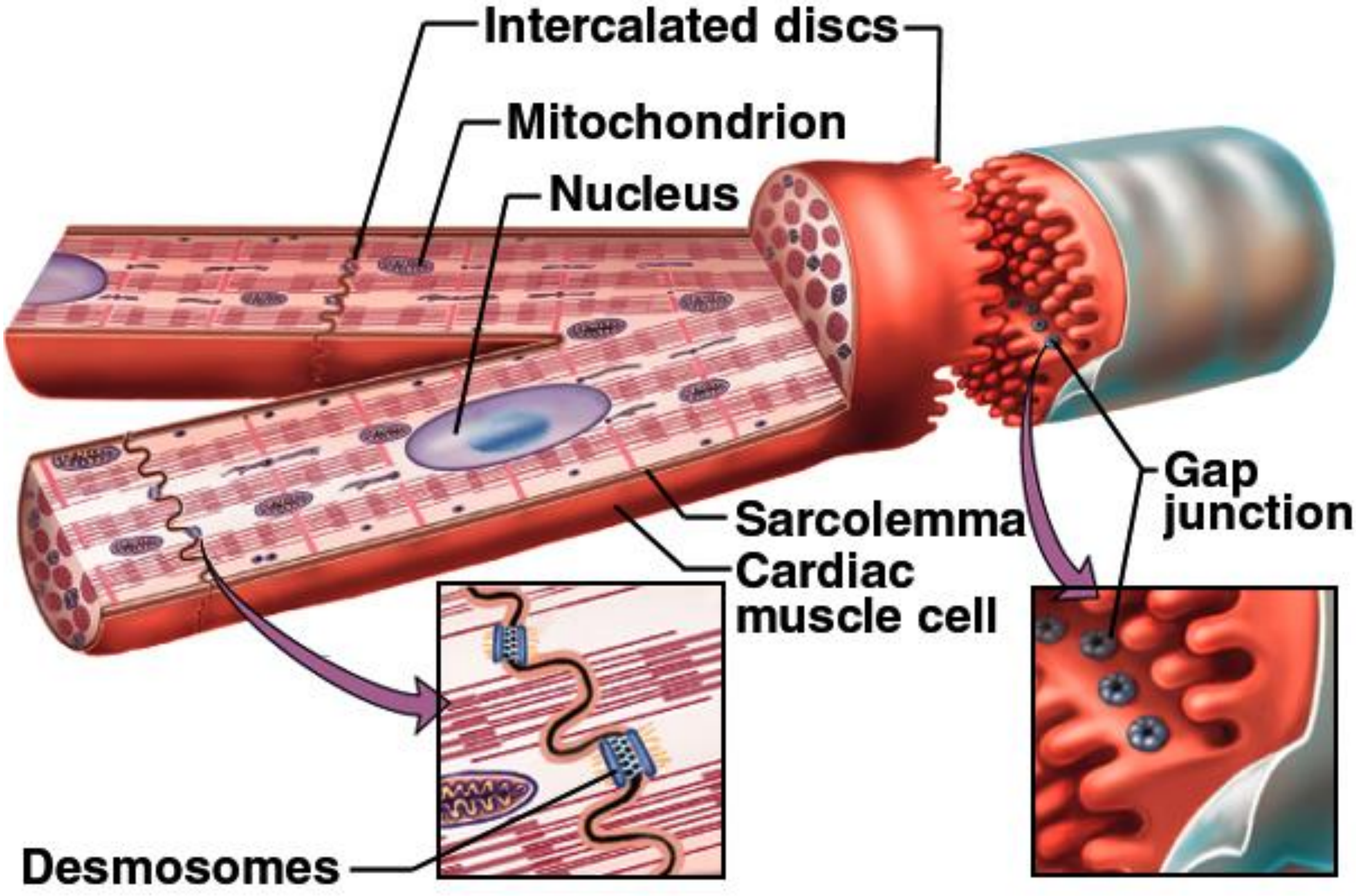
Venous Drainage

- 20% drains directly into right ventricle
- 80% returns to right atrium
  - – great cardiac vein (anterior interventricular sulcus)
  - – middle cardiac vein (posterior sulcus)
  - – coronary sinus (posterior coronary sulcus) collects blood from these and smaller veins and empties into right atrium

Coronary Flow and the Cardiac Cycle

- Reduced during ventricular contraction
  - arteries compressed
- Increased during ventricular relaxation
  - openings to coronary arteries, just above aortic semilunar valve, fill as blood surges back to valve

Structure of Cardiac Muscle Cell

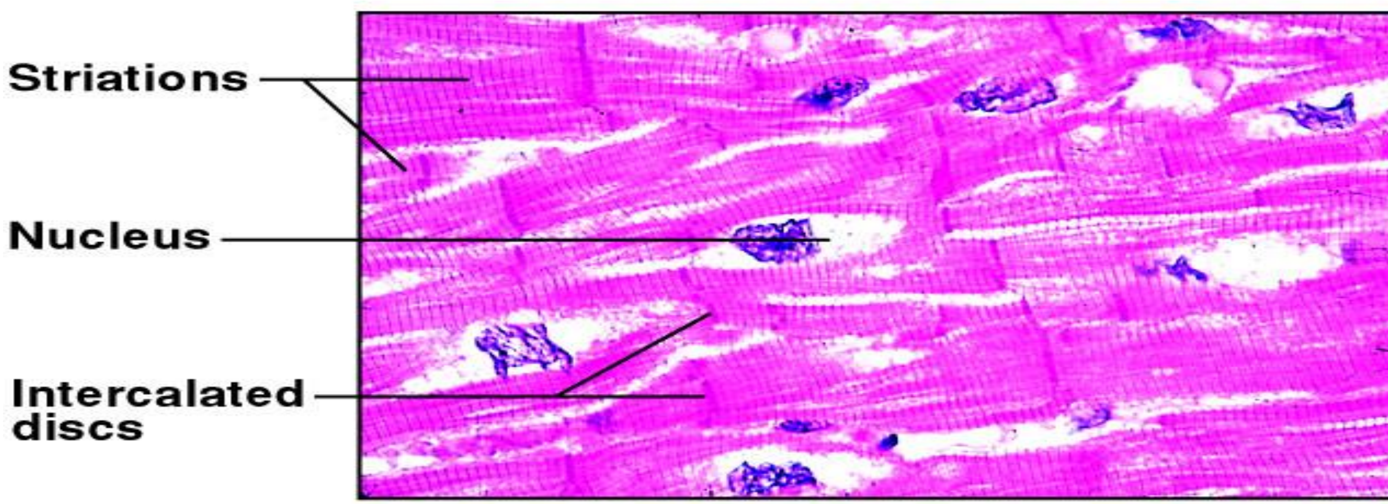


Structure of Cardiac Muscle

- Short, thick, branched cells, 50 to 100 μm long and 10 to 20 μm wide with one central nucleus
- $\downarrow$ Sarcoplasmic reticulum, large T tubules
  - must admit more Ca^{2+} from ECF during excitation
- Intercalated discs, join myocytes end to end
  - interdigitating folds - $\uparrow$ surface area
  - mechanical junctions tightly join myocytes
    - fascia adherens: actin anchored to plasma membrane
    - desmosomes
  - electrical junctions - gap junctions form channels allowing ions to flow directly into next cell

Metabolism of Cardiac Muscle

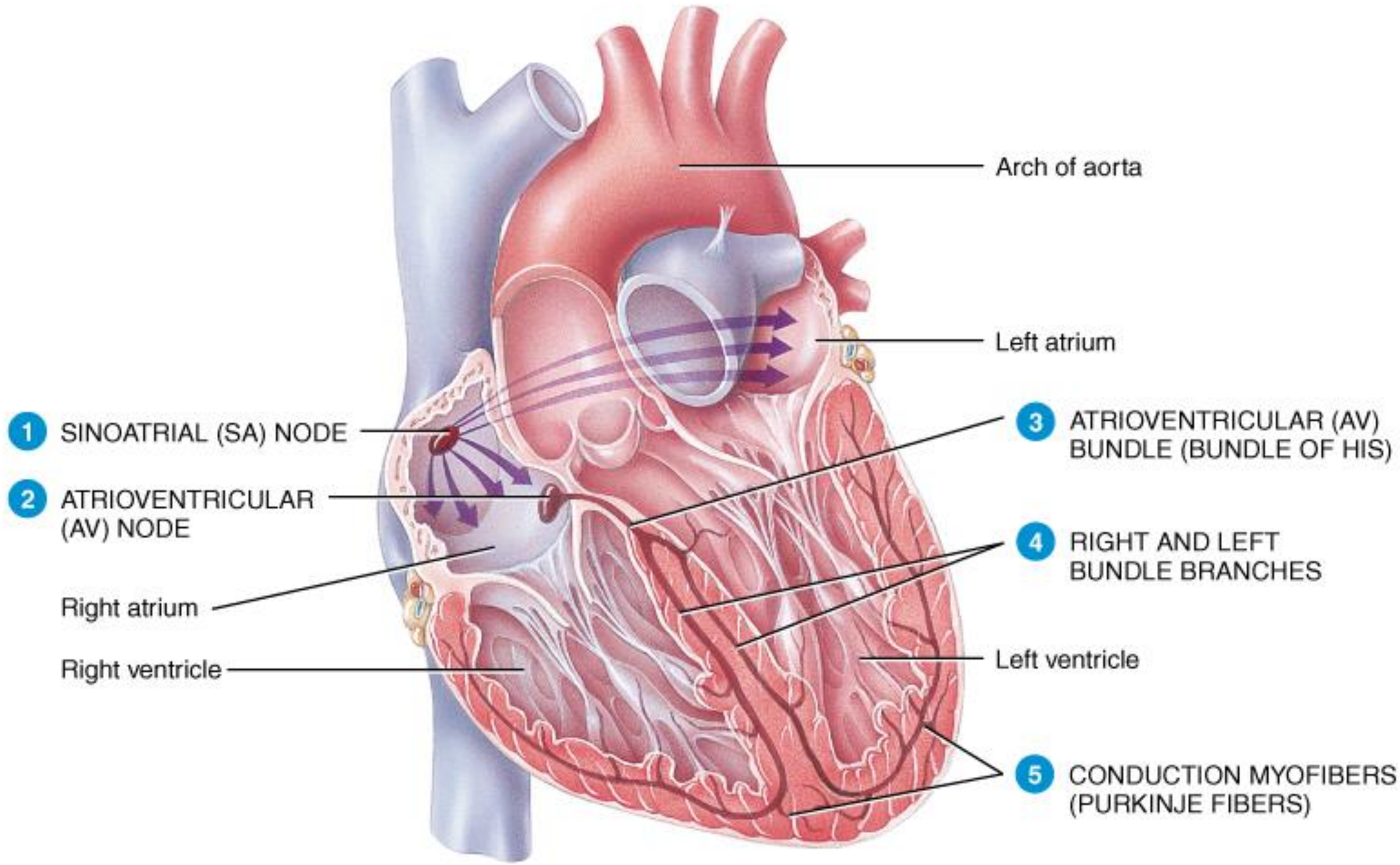
- Aerobic respiration
- Rich in myoglobin and glycogen
- Large mitochondria
- Organic fuels: fatty acids, glucose, ketones
- Fatigue resistant



Cardiac Conduction System

- Myogenic (automaticity) - heartbeat originates within heart
- Autorhythmic – regular, spontaneous depolarization
- Conduction system
  - SA node: pacemaker, initiates heartbeat, sets heart rate
  - *fibrous skeleton insulates atria from ventricles*
  - AV node: electrical gateway to ventricles
  - AV bundle of His: pathway for signals from AV node
  - Right and left bundle branches: divisions of AV bundle that enter interventricular septum and descend to apex
  - Purkinje fibers: upward from apex spread throughout ventricular myocardium

Conduction System of Heart



Coordinates contraction of heart muscle.

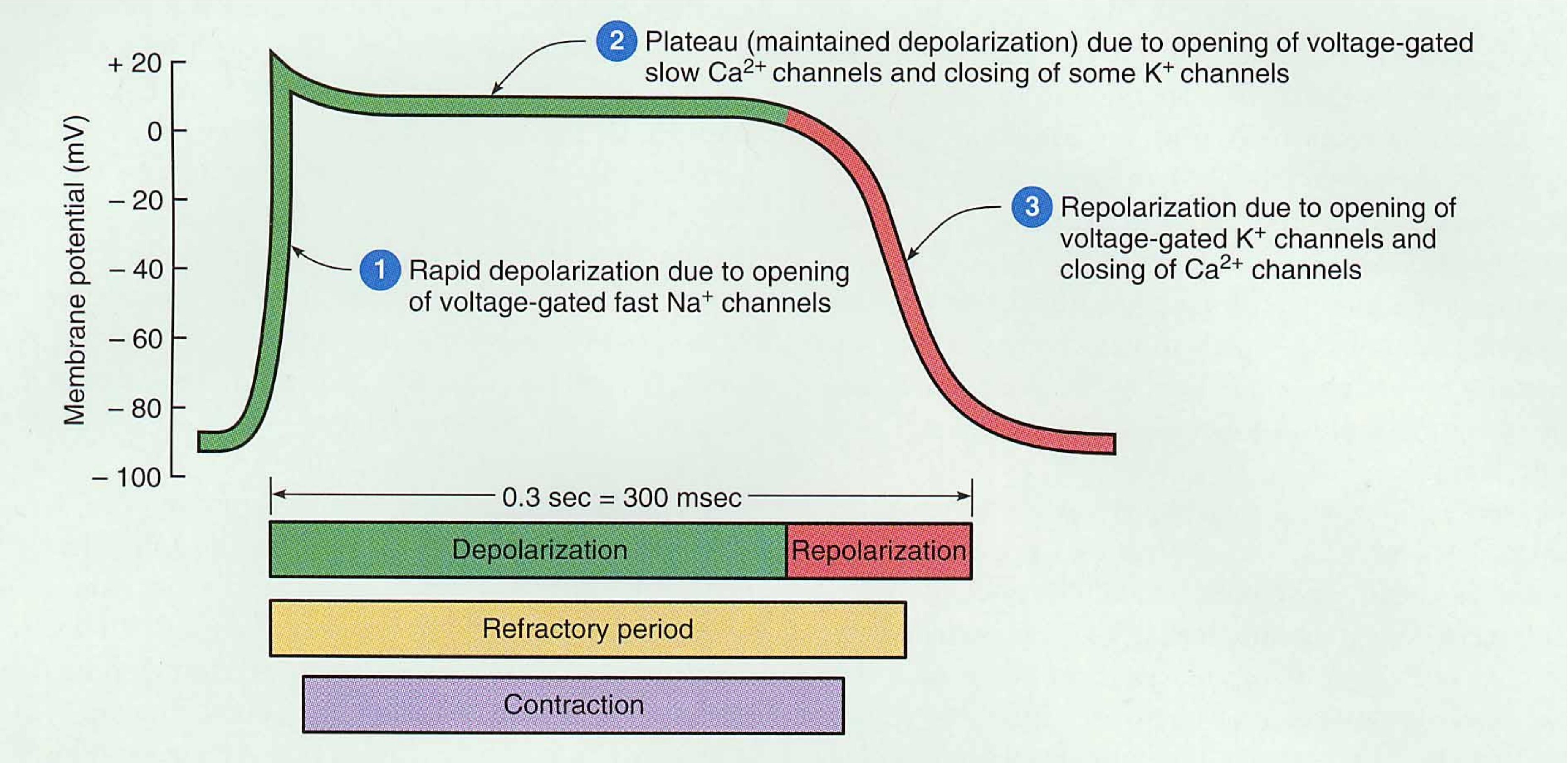
Rhythm of Conduction System

- Systole=contraction;   Diastole=relaxation
- SA node fires spontaneously 90-100 times per minute
- AV node fires at 40-50 times per minute
- If both nodes are suppressed fibers in ventricles by themselves fire only 20-40 times per minute
- Artificial pacemaker needed if pace is too slow
- Extra beats forming at other sites are called ectopic pacemakers
  - caffeine & nicotine increase activity

Timing of Atrial & Ventricular Excitation

- SA node setting pace since is the fastest
- In 50 msec excitation spreads through both atria and down to AV node
- 100 msec delay at AV node due to smaller diameter fibers- allows atria to fully contract filling ventricles before ventricles contract
- In 50 msec excitation spreads through both ventricles simultaneously

Physiology of Contraction

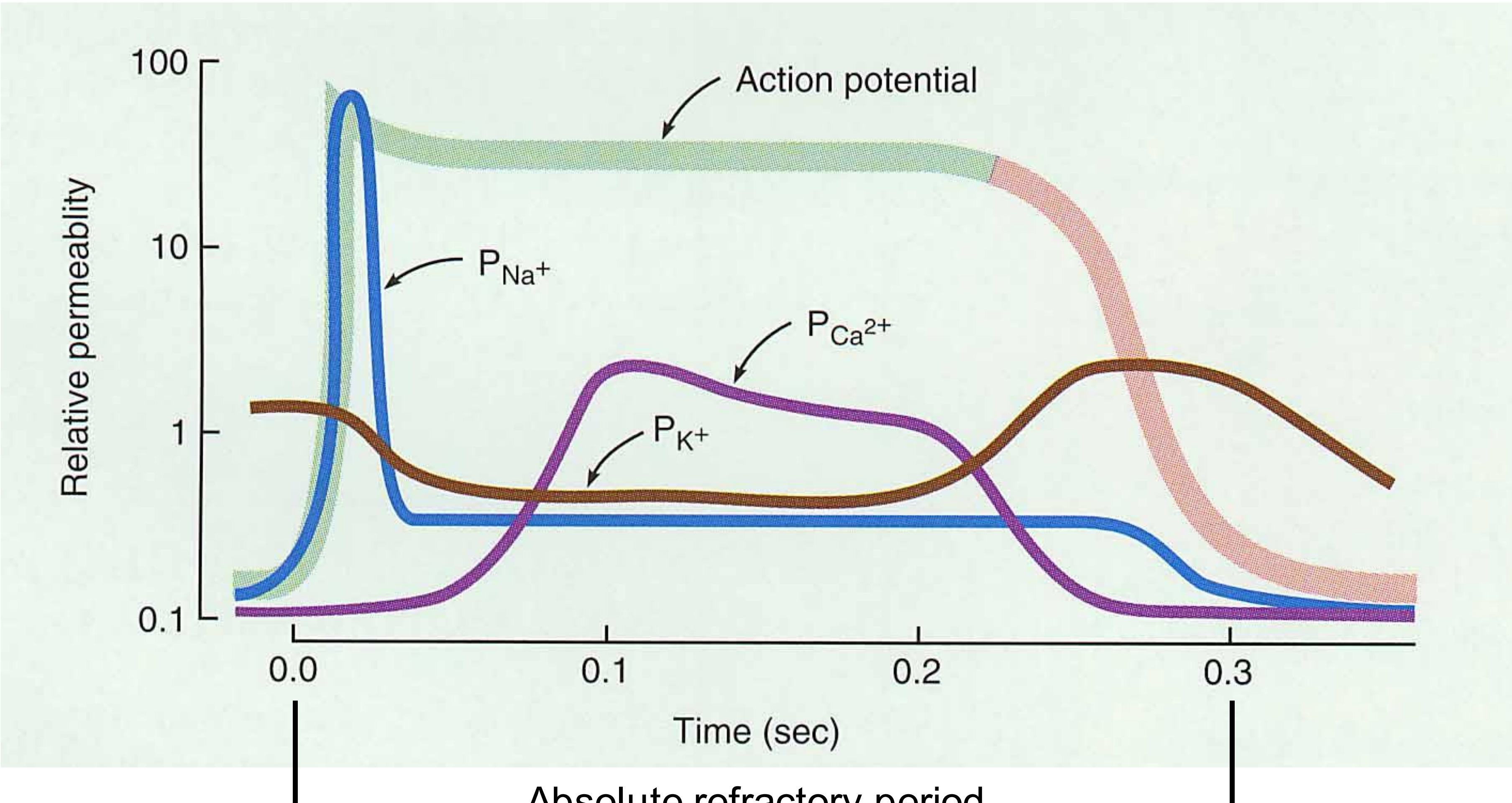


- Depolarization, plateau, repolarization

Depolarization & Repolarization

- Depolarization
  - Cardiac cell resting membrane potential is -90mv
  - excitation spreads through gap junctions
  - fast Na+ channels open for rapid depolarization
- Plateau phase
  - 250 msec (only 1msec in neuron)
  - slow Ca^{+2} channels open, let Ca^{+2} enter from outside cell and from storage in sarcoplasmic reticulum, while K^+ channels close
  - Ca^{+2} binds to troponin to allow for actin-myosin cross-bridge formation & tension development
- Repolarization
  - Ca^{+2} channels close and K^+ channels open & -90mv is restored as potassium leaves the cell
- Refractory period
  - very long so heart can fill

Action Potential in Cardiac Muscle



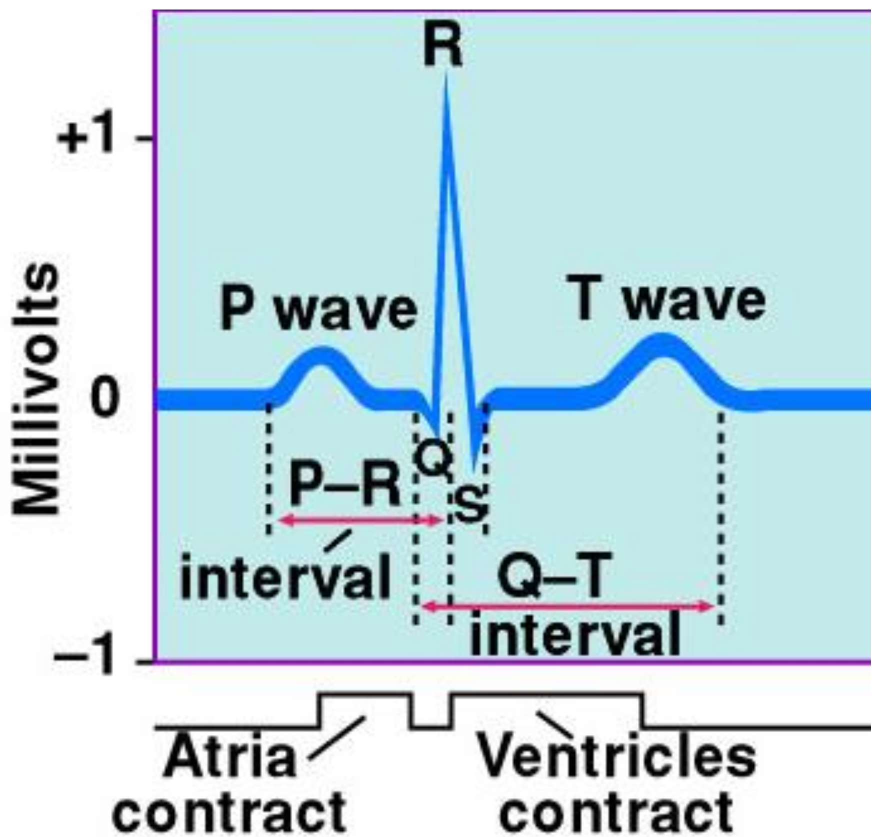
Changes in cell membrane permeability.

Contraction of Myocardium

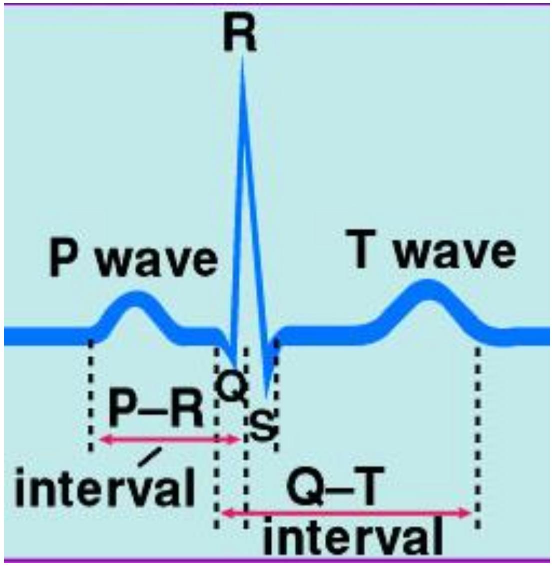
- Myocytes have stable resting potential of -90 mV
- Depolarization (very brief)
  - stimulus opens voltage regulated Na^+ gates, (Na^+ rushes in) membrane depolarizes rapidly
  - action potential peaks at +30 mV
  - Na^+ gates close quickly
- Plateau - 200 to 250 msec, sustains contraction
  - slow Ca^{+2} channels open, Ca^{+2} binds to fast Ca^{+2} channels on SR, releases ↑Ca^{+2} into cytosol: contraction
- Repolarization - Ca^{+2} channels close, K^+ channels open, rapid K^+ out returns to resting potential

Electrocardiogram (ECG)

- Composite of all action potentials of nodal and myocardial cells detected, amplified and recorded by electrodes on arms, legs and chest

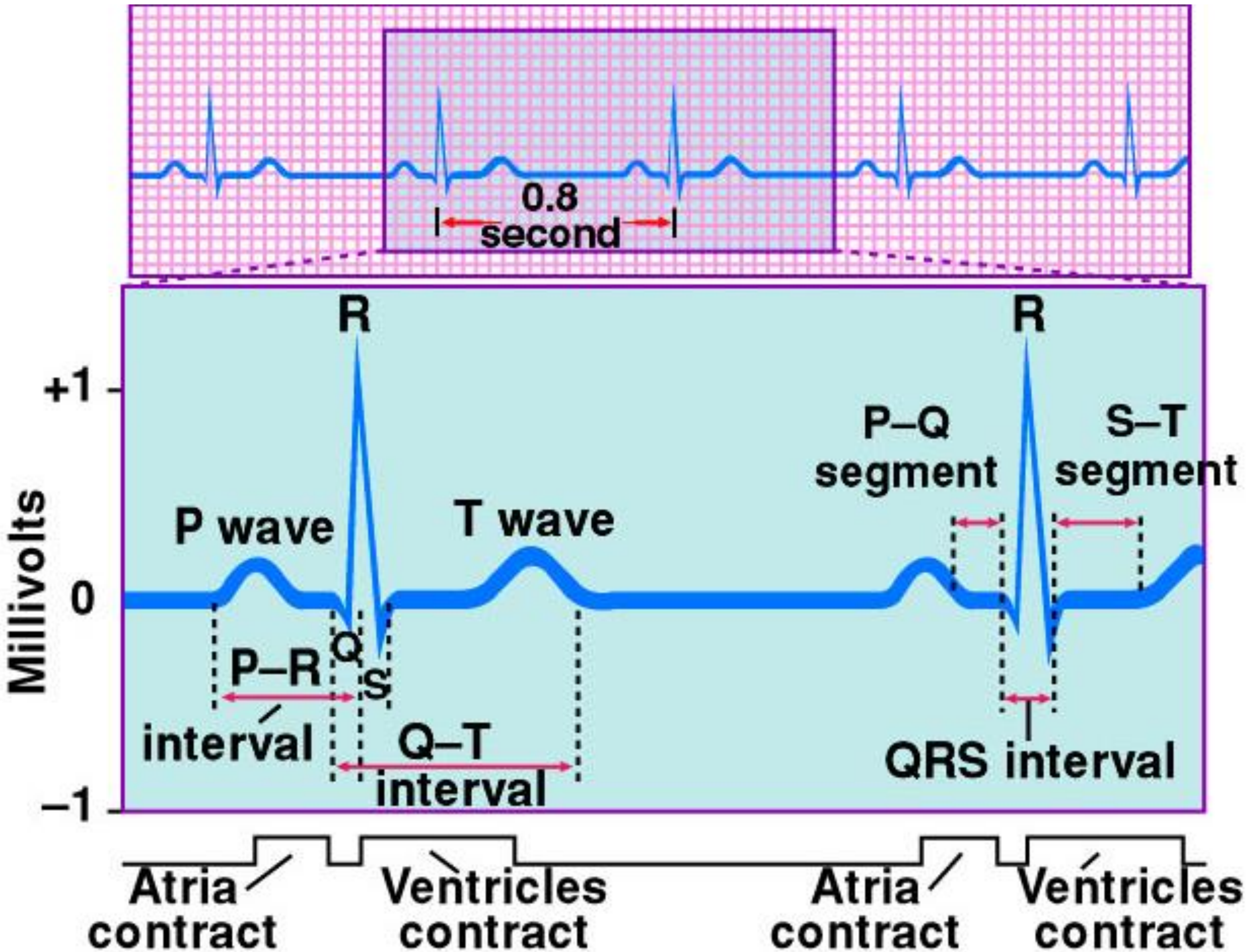


ECG



- P wave
  - SA node fires, atrial depolarization
  - atrial systole
- PQ interval
  - Conduction time from atrial to ventricular excitation
- QRS complex
  - AV node fires, ventricular depolarization
  - ventricular systole
  - (atrial repolarization and diastole - signal obscured)
- T wave
  - ventricular repolarization

Normal Electrocardiogram (ECG)



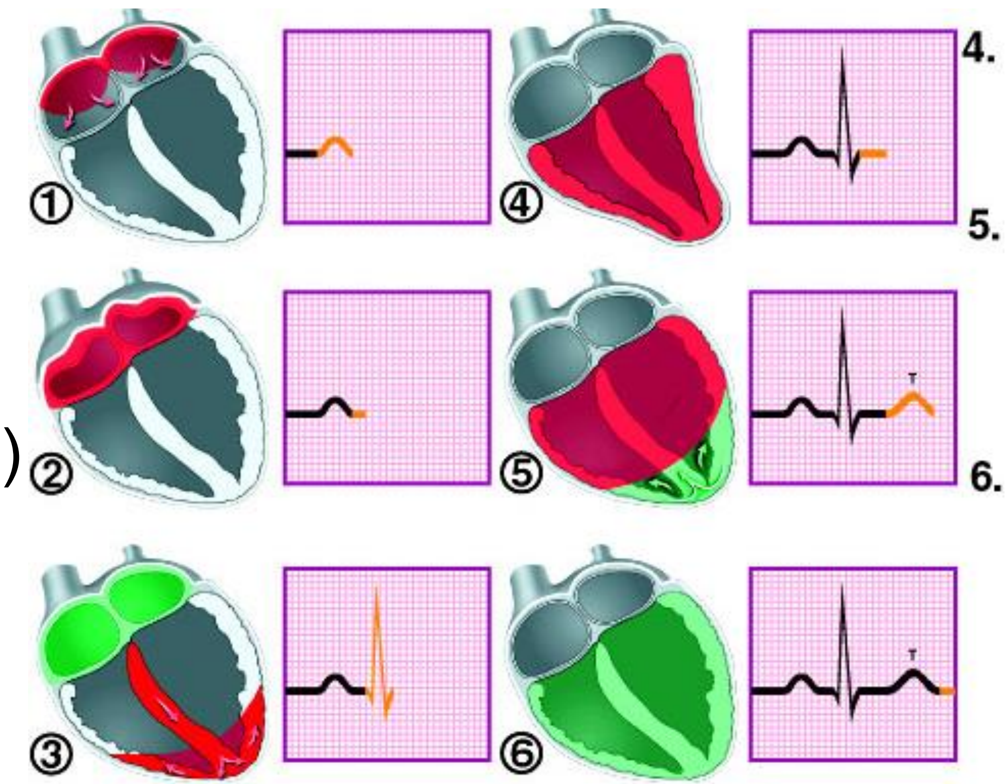
Electrical Activity of Myocardium

1) **atrial** depolarization begins

2) atrial depolarization complete (atria contracted)

3) **ventricles** begin to depolarize at apex; atria repolarize (atria relaxed)

4) ventricular depolarization complete (ventricles contracted)



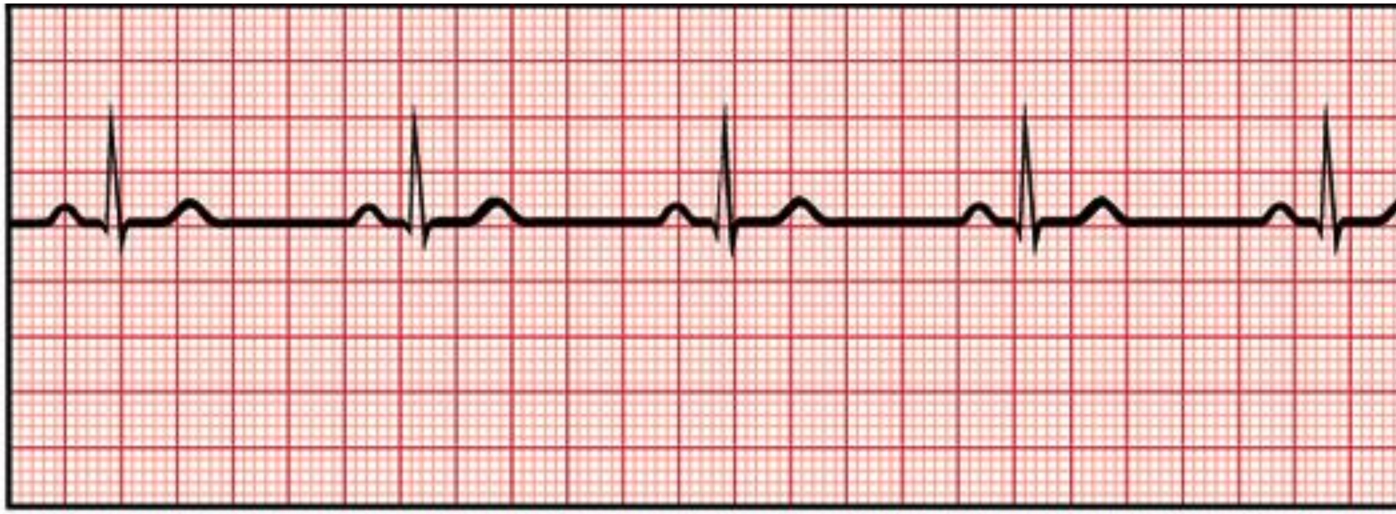
5) ventricles begin to repolarize at apex

6) ventricular repolarization complete (ventricles relaxed)

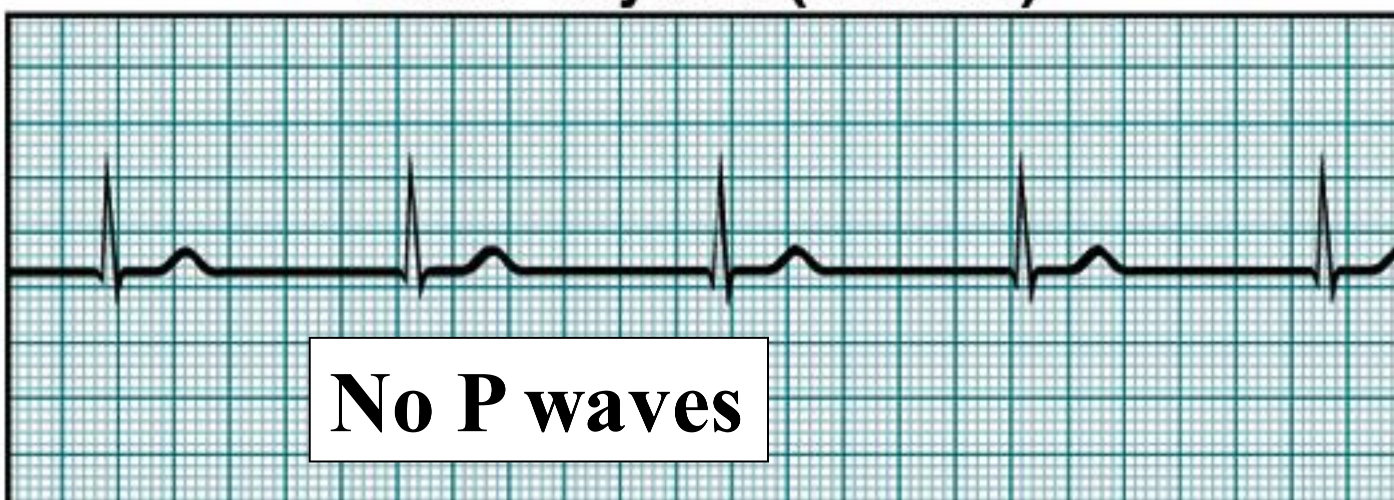
Diagnostic Value of ECG

- Invaluable for diagnosing abnormalities in conduction pathways, MI, heart enlargement and electrolyte and hormone imbalances

ECGs, Normal & Abnormal

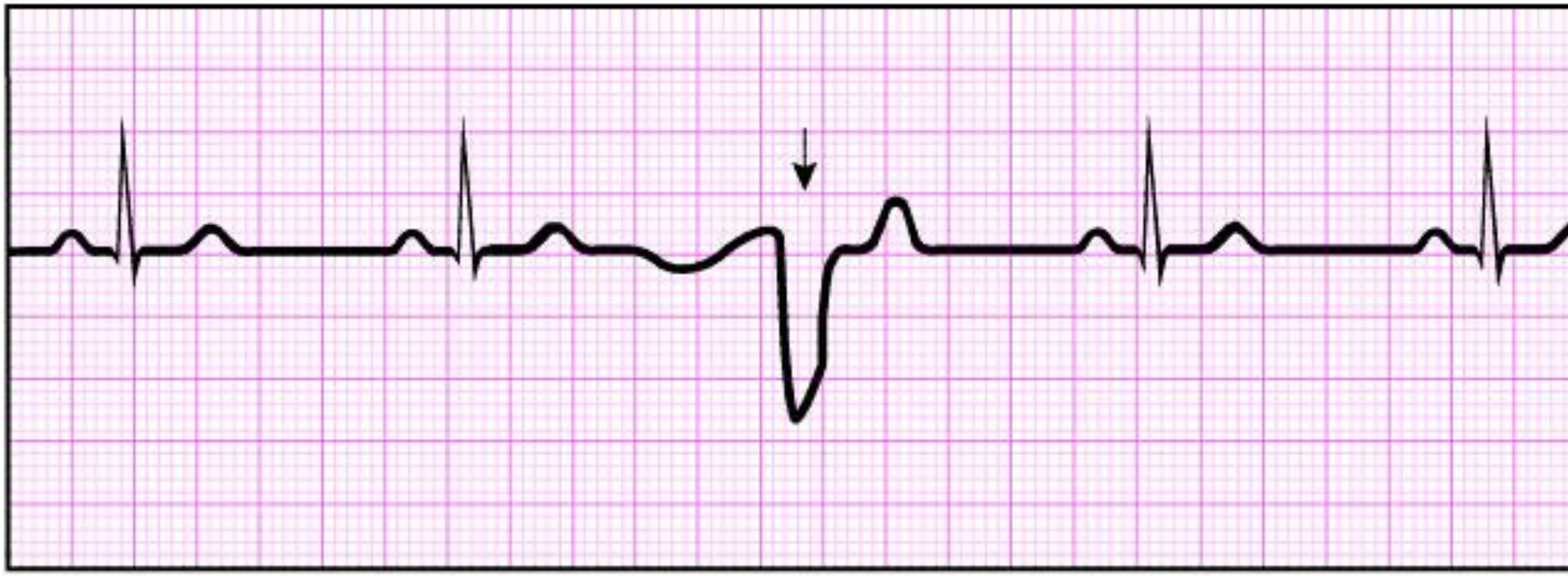


Sinus rhythm (normal)



Nodal rhythm – no SA node activity

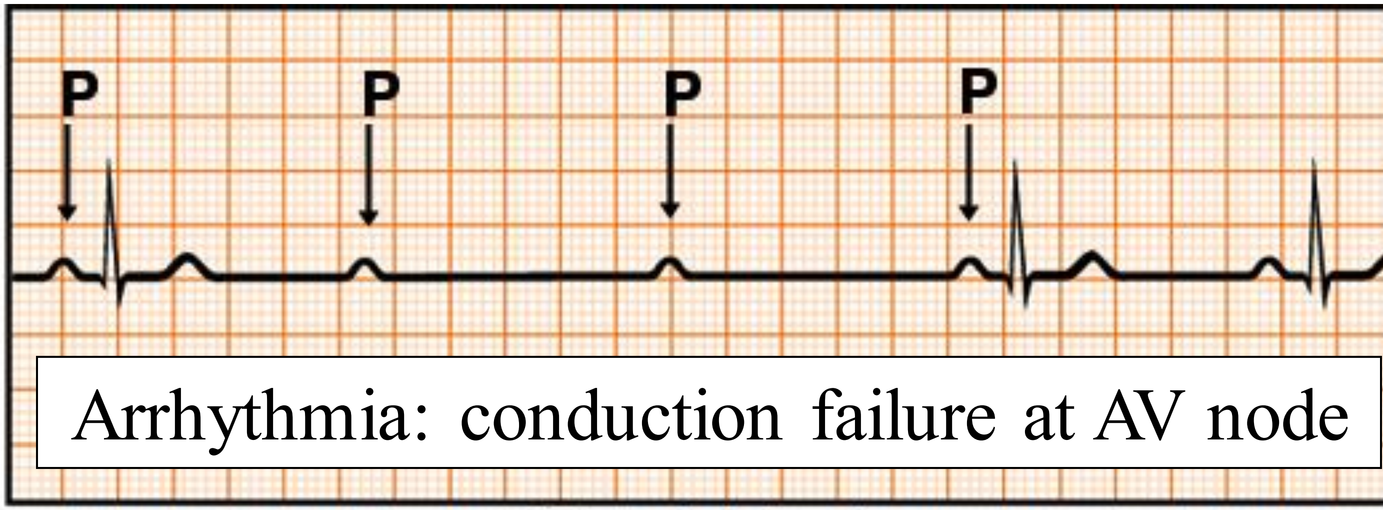
ECGs, Abnormal



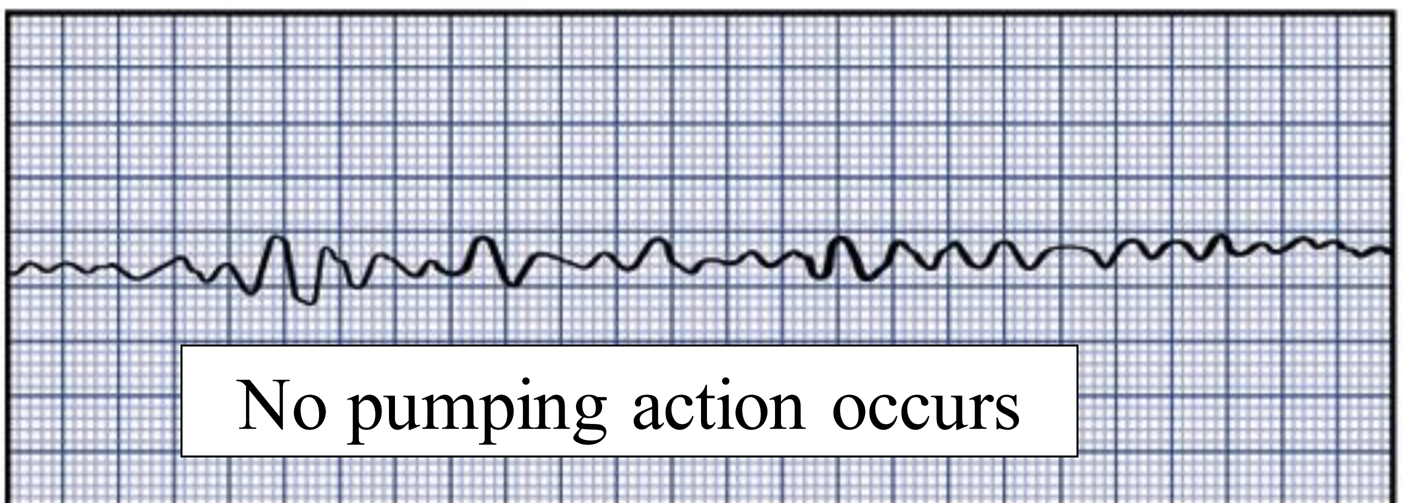
Premature ventricular contraction

Extrasystole : note the inverted QRS complex, misshapen QRS and T and absence of a P wave preceding this contraction.

ECGs, Abnormal



Heart block



Fibrillation

Rate of Cardiac Cycle

- Atrial systole, 0.1 sec
- Ventricular systole, 0.3 sec
- Quiescent period, 0.4 sec
- Total 0.8 sec, heart rate 75 bpm

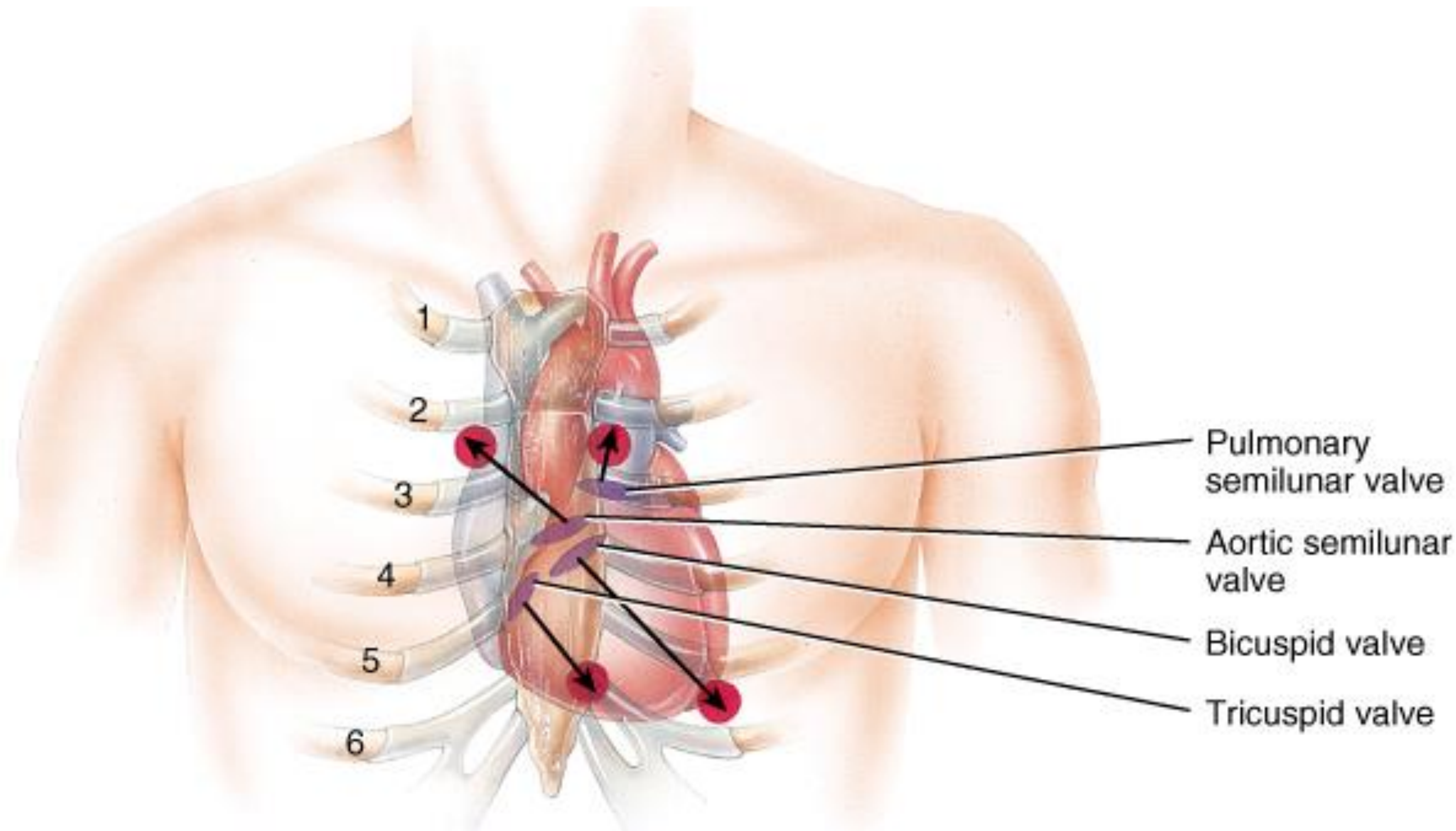
One Cardiac Cycle

- At 75 beats/min, one cycle requires 0.8 sec.
  - systole (contraction) and diastole (relaxation) of both atria, plus the systole and diastole of both ventricles
- End diastolic volume (EDV)
  - volume in ventricle at end of diastole, about 130ml
- End systolic volume (ESV)
  - volume in ventricle at end of systole, about 60ml
- Stroke volume (SV)
  - the volume ejected per beat from each ventricle, about 70ml
  - SV = EDV - ESV

Heart Sounds

- Auscultation - listening to sounds made by body
- First heart sound (S1), louder and longer “lubb”, occurs with closure of AV valves
- Second heart sound (S2), softer and sharper “dupp” occurs with closure of semilunar valves
- S3 - rarely heard in people > 30

Heart Sounds



Where to listen on chest wall for heart sounds.

Phases of Cardiac Cycle

- Quiescent period
  - all chambers relaxed
  - AV valves open
  - blood flowing into ventricles
- Atrial systole
  - SA node fires, atria depolarize
  - P wave appears on ECG
  - atria contract, force additional blood into ventricles
  - ventricles now contain end-diastolic volume (EDV) of about 130 ml of blood

Isovolumetric Contraction of Ventricles

- Atria repolarize and relax
- Ventricles depolarize
- QRS complex appears in ECG
- Ventricles contract
- Rising pressure closes AV valves - heart sound S1 occurs
- No ejection of blood yet (no change in volume)

Ventricular Ejection

- Rising pressure opens semilunar valves
- Rapid ejection of blood
- Reduced ejection of blood (less pressure)
- Stroke volume: amount ejected, 70 ml at rest
- SV/EDV= ejection fraction, at rest ~ 54%, during vigorous exercise as high as 90%, diseased heart < 50%
- End-systolic volume: amount left in heart

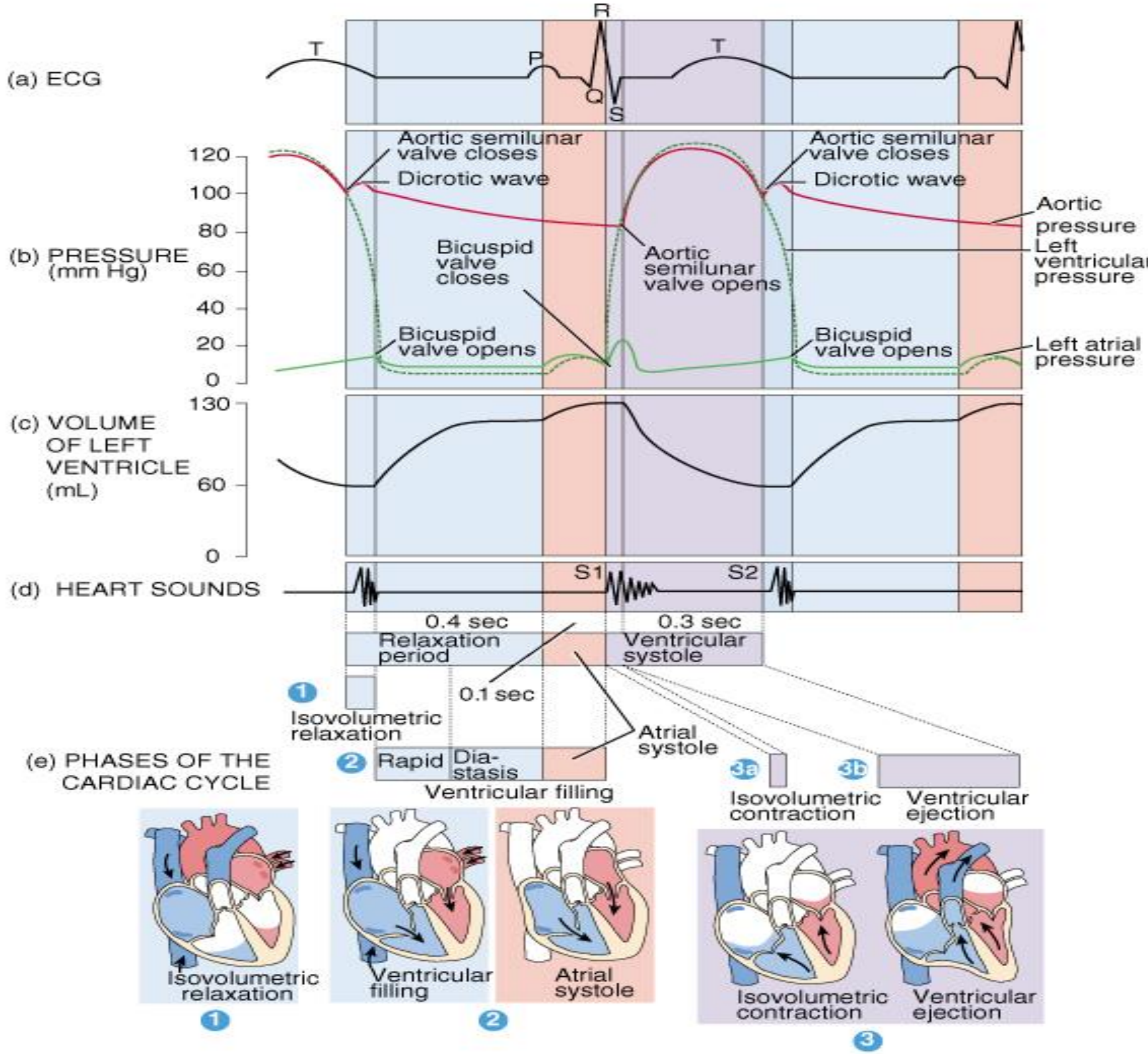
Isovolumetric Relaxation of Ventricles

- T wave appears in ECG
- Ventricles repolarize and relax (begin to expand)
- Semilunar valves close (dicrotic notch of aortic press. curve) - heart sound S2 occurs
- AV valves remain closed
- Ventricles expand but do not fill (no change in volume)

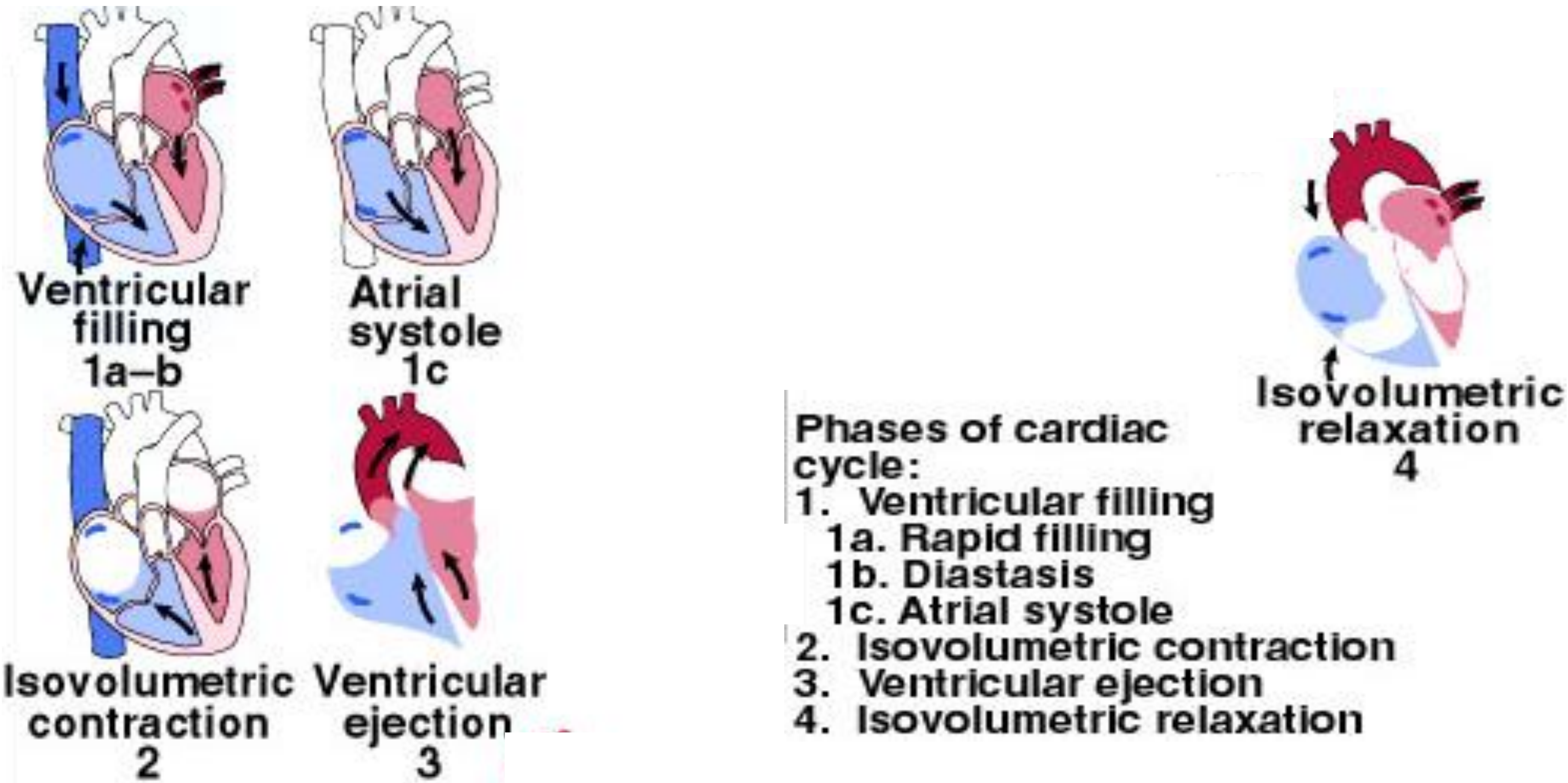
Ventricular Filling - 3 phases

- Rapid ventricular filling - AV valves first open ↑pressure
- Diastasis - sustained lower pressure, venous return
- Filling completed by atrial systole
- Heart sound S3 may occur

Cardiac Cycle



Events of the Cardiac Cycle



Rate of Cardiac Cycle

- Atrial systole, 0.1 sec
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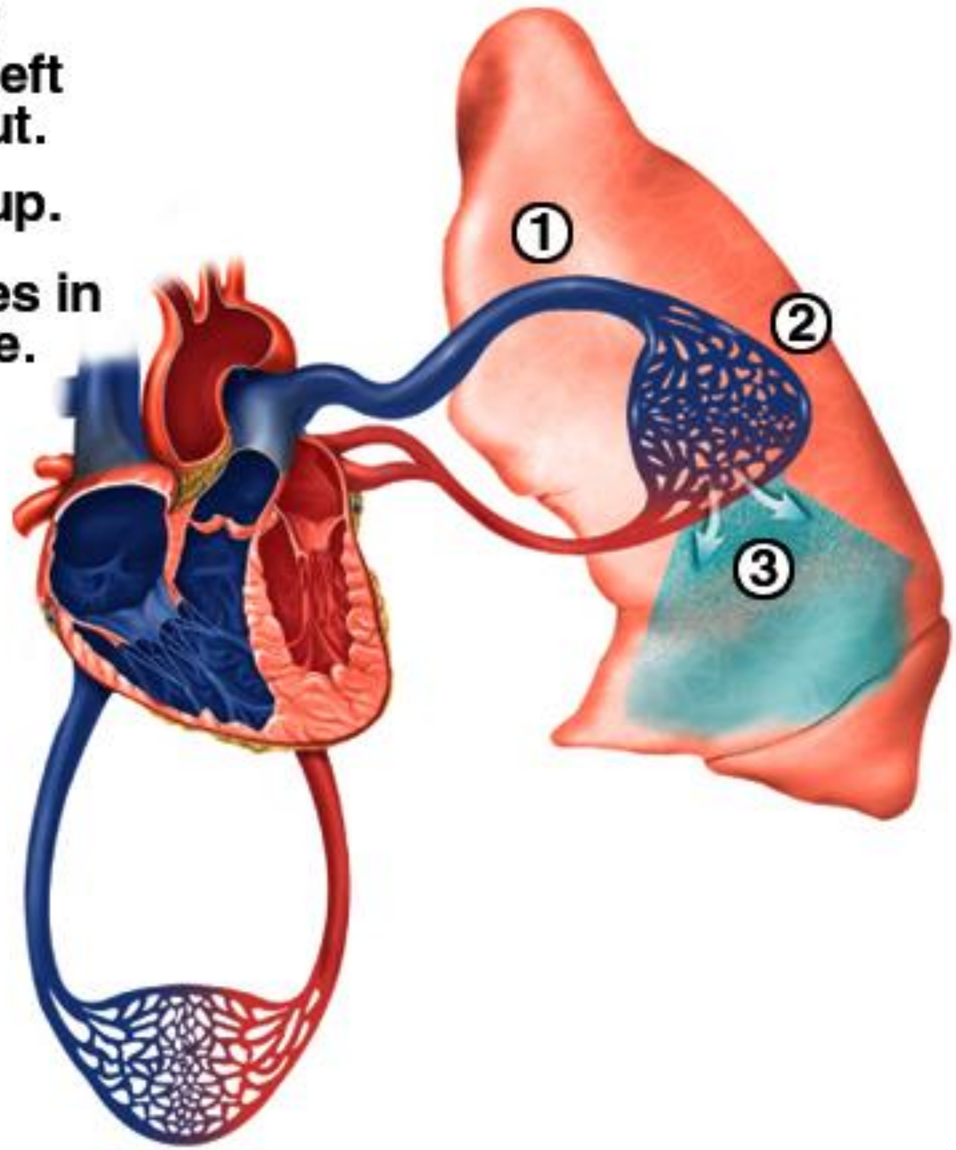
- Quiescent period, 0.4 sec
- Total 0.8 sec, heart rate 75 bpm

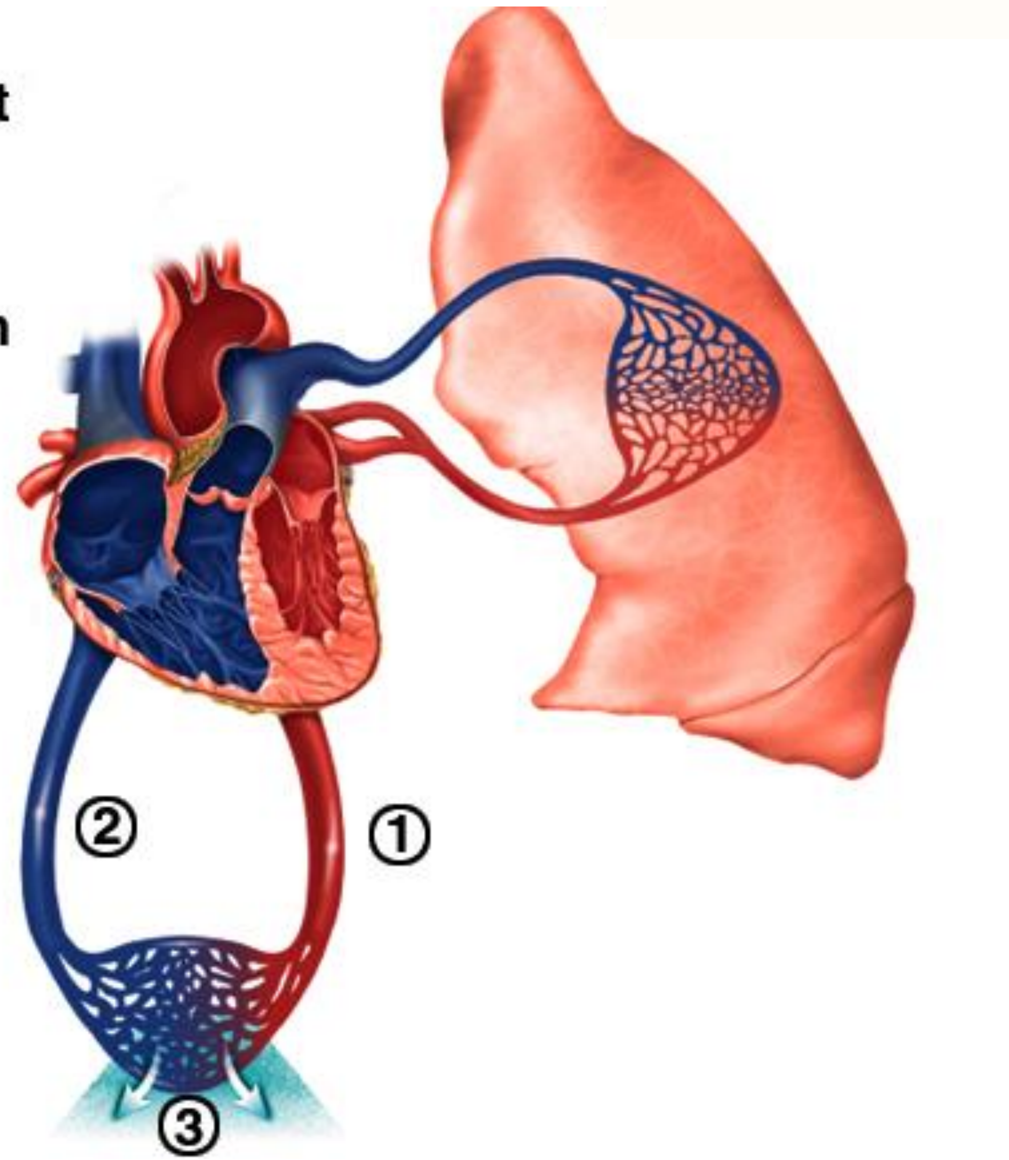
Unbalanced Ventricular Output

1. Right ventricular output exceeds left ventricular output.
2. Pressure backs up.
3. Fluid accumulates in pulmonary tissue.



Unbalanced Ventricular Output

1. Left ventricular output exceeds right ventricular output.
2. Pressure backs up.
3. Fluid accumulates in systemic tissue.



Congestive Heart Failure

- Causes of CHF
  - coronary artery disease, hypertension, MI, valve disorders, congenital defects
- Left side heart failure
  - less effective pump so more blood remains in ventricle
  - heart is overstretched & even more blood remains
  - blood backs up into lungs as pulmonary edema
  - suffocation & lack of oxygen to the tissues
- Right side failure
  - fluid builds up in tissues as peripheral edema

Cardiac Output (CO)

- Amount ejected by each ventricle in 1 minute
- CO = HR x SV
- Resting values, CO = 75 beats/min x70 ml/beat = 5,250 ml/min, usually about 4 to 6L/min
- Vigorous exercise ↑ CO to 21 L/min for fit person and up to 35 L/min for world class athlete
- Cardiac reserve: Maximum output/output at rest
  - ↑ with fitness, ↓ with disease
  - Average 4-5, athlete 7-8

Heart Rate

- Measured from pulse
- Infants have HR of 120 beats per minute or more
- Young adult females avg. 72 - 80 bpm
- Young adult males avg. 64 to 72 bpm
- HR rises again in the elderly
- Tachycardia: persistent, resting adult HR > 100
  - stress, anxiety, drugs, heart disease or ↑ body temp.
- Bradycardia: persistent, resting adult HR < 60
  - common in sleep and endurance trained athletes (↑ SV)

Chronotropic Effects

- Positive chronotropic agents ↑ HR and negative chronotropic agents ↓ HR
- Cardiac center of medulla oblongata
  - an autonomic control center with 2 neuronal pools: a cardioacceleratory center (sympathetic), and a cardioinhibitory center (parasympathetic)

Sympathetic Nervous System

- Cardioacceleratory center of Medulla
  - stimulates sympathetic cardiac accelerator nerves to SA node, AV node and myocardium
  - these nerves secrete norepinephrine, which binds to β-adrenergic receptors in the heart (+ chronotropic effect)
  - CO peaks at HR of 160 to 180 bpm
  - Sympathetic n.s. can $\uparrow$ HR up to 230 bpm, (limited by refractory period of SA node), but SV and CO $\downarrow$ (less filling time)

Parasympathetic Nervous System

- Cardioinhibitory center of medulla
  - stimulates vagus nerves
    - right vagus nerve - SA node
    - left vagus nerve - AV node
  - secrete ACH (acetylcholine), binds to muscarinic receptors
    - opens K^+ channels in nodal cells, hyperpolarized, fire less frequently, HR slows down
  - vagal tone: background firing rate holds HR to sinus rhythm of 70 to 80 bpm
    - severed vagus nerves - SA node fires at intrinsic rate-100bpm
    - maximum vagal stimulation ↓ HR as low as 20 bpm

Inputs to Cardiac Center

- Higher brain centers affect HR
  - sensory and emotional stimuli - rollercoaster, IRS audit
  - cerebral cortex, limbic system, hypothalamus
- Proprioceptors
  - inform cardiac center about changes in activity, HR ↑ before metabolic demands arise
- Baroreceptors
  - pressure sensors in aorta and internal carotid arteries send continual stream of signals to cardiac center
    - if pressure drops, signal rate drops, cardiac center ↑ HR
    - if pressure rises, signal rate rises, cardiac center ↓ HR

Inputs to Cardiac Center 2

- Chemoreceptors
  - sensitive to blood pH, CO_2 and oxygen
  - aortic arch, carotid arteries and medulla oblongata
  - primarily respiratory control, may influence HR
  - $\uparrow CO_2$ (hypercapnia) causes $\uparrow H^+$ levels, may create acidosis ($pH < 7.35$)
  - Hypercapnia and acidosis stimulates cardiac center to $\uparrow$ HR

Chronotropic Chemicals

- Neurotransmitters - with cAMP as 2 messenger
  - norepinephrine and epinephrine (catecholamines) are potent cardiac stimulants
- Drugs
  - caffeine inhibits cAMP breakdown
  - nicotine stimulates catecholamine secretion
- Hormones
  - TH ↑ adrenergic receptors in heart, ↑ sensitivity to sympathetic stimulation, ↑ HR
- Electrolytes - K has greatest chronotropic effects
  - ↑ K myocardium less excitable, HR slow and irregular
  - ↓ K cells hyperpolarized, requires ↑ stimulation

Stroke Volume

- Governed by three factors:
  - preload, contractility and afterload
- ↑ preload or contractility ↑ SV
- ↑ afterload ↓ SV

Preload

- Amount of tension in ventricular myocardium before it contracts
- ↑ preload causes ↑ contraction strength
  - exercise ↑ venous return, stretches myocardium
    (↑ preload) , myocytes generate more tension during contraction, ↑ CO matches ↑ venous return
- Frank-Starling law of heart - SV $\propto$ EDV
  - ventricles eject as much blood as they receive; more they are stretched *(↑ preload)* the harder they contract

Contractility

- Contraction force for a **given** preload
- Tension caused by factors that adjust myocyte's responsiveness to stimulation
  - factors that ↑ contractility are positive inotropic agents
    - NE, hypercalcemia, catecholamines, glucagon, digitalis
  - factors that ↓ contractility are negative inotropic agents
    - hyperkalemia (K^+), hypocalcemia, hypoxia, hypercapnia

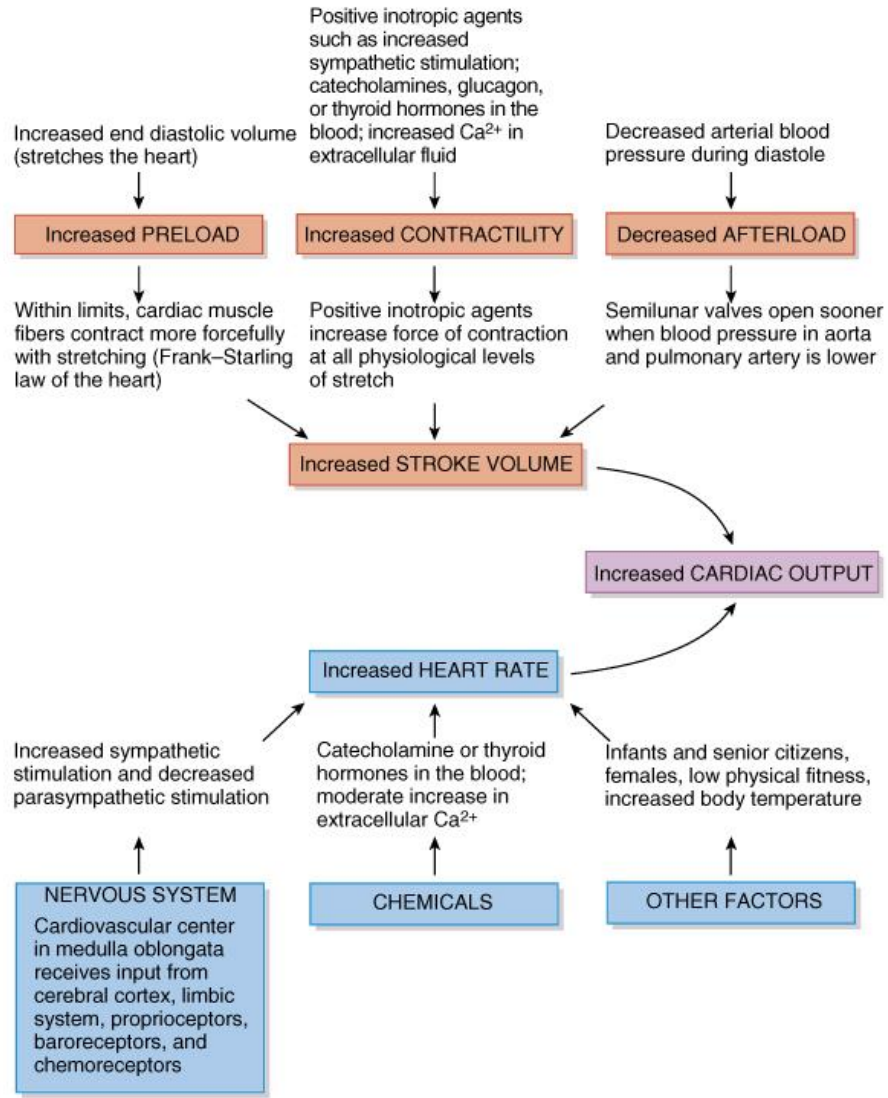
Afterload

- Pressure in arteries above semilunar valves opposes opening of valves
- ↑ afterload ↓ SV
  - any impedance in arterial circulation ↑ afterload
- Continuous ↑ in afterload (lung disease, atherosclerosis, etc.) causes hypertrophy of myocardium, may lead it to weaken and fail

Exercise and Cardiac Output

- Effect of proprioceptors
  - HR ↑ at beginning of exercise due to signals from joints, muscles
- Effect of venous return
  - muscular activity ↑ venous return causes ↑ SV
- ↑ HR and ↑ SV cause ↑CO
- Effect of ventricular hypertrophy
  - caused by sustained program of exercise
  - ↑ SV allows heart to beat more slowly at rest, **40-60bpm**
  - ↑ cardiac reserve, can tolerate more exertion

Stroke Volume and Heart Rate



Risk Factors for Heart Disease

- Risk factors in heart disease:
  - high blood cholesterol level
  - high blood pressure
  - cigarette smoking
  - obesity & lack of regular exercise.
- Other factors include:
  - diabetes mellitus
  - genetic predisposition
  - male gender
  - high blood levels of fibrinogen
  - left ventricular hypertrophy

Plasma Lipids and Heart Disease

- Risk factor for developing heart disease is high blood cholesterol level.
  - promotes growth of fatty plaques
  - Most lipids are transported as lipoproteins
    - low-density lipoproteins (LDLs)
    - high-density lipoproteins (HDLs)
    - very low-density lipoproteins (VLDLs)
  - HDLs remove excess cholesterol from circulation
  - LDLs are associated with the formation of fatty plaques
  - VLDLs contribute to increased fatty plaque formation
- There are two sources of cholesterol in the body:
  - in foods we ingest & formed by liver

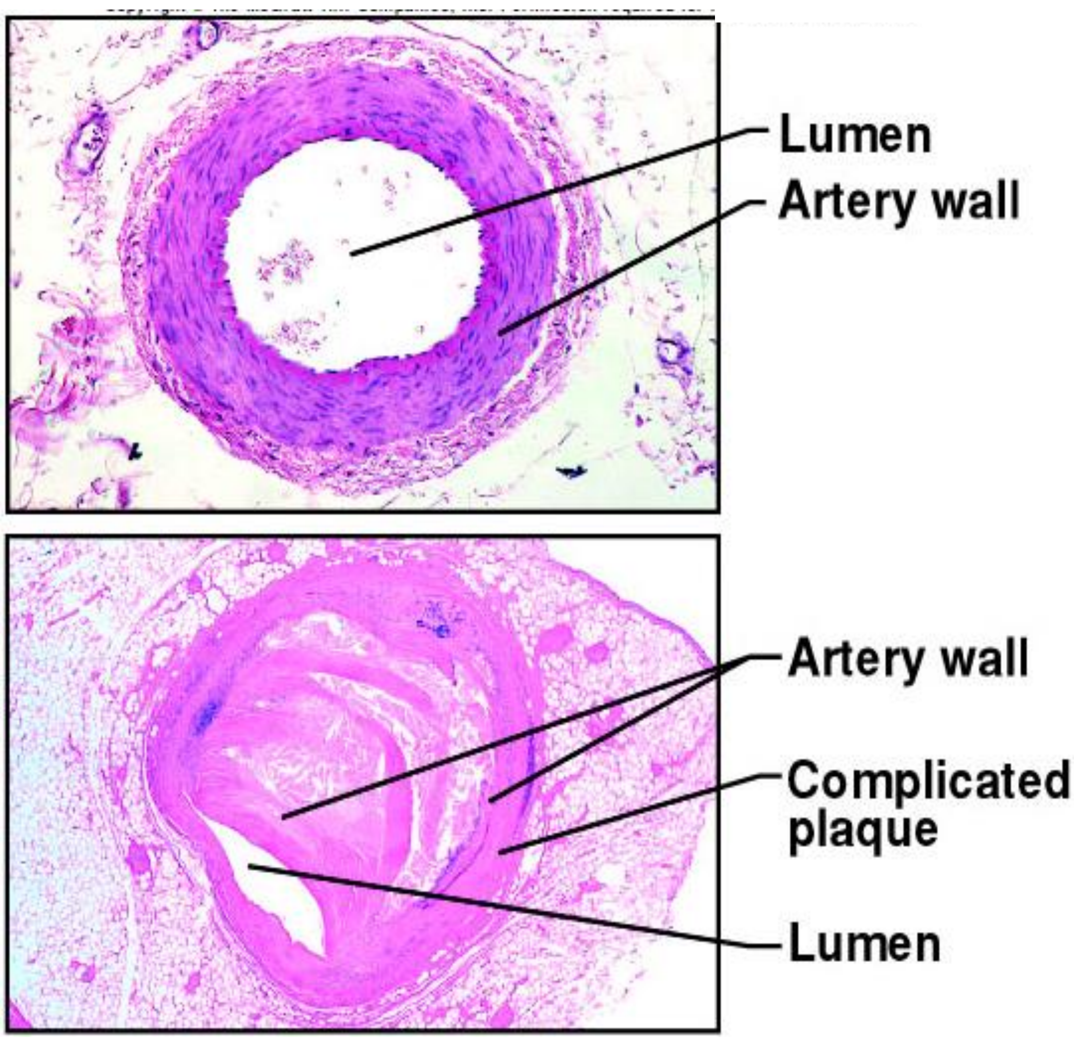
Desirable Levels of Blood Cholesterol for Adults

- TC (total cholesterol) under 200 mg/dl
- LDL under 130 mg/dl
- HDL over 40 mg/dl
- Normally, triglycerides are in the range of 10-190 mg/dl.
- Among the therapies used to reduce blood cholesterol level are exercise, diet, and drugs.

Exercise and the Heart

- Sustained exercise increases oxygen demand in muscles.
- Benefits of aerobic exercise (any activity that works large body muscles for at least 20 minutes, preferably 3-5 times per week) are;
  - increased cardiac output
  - increased HDL and decreased triglycerides
  - improved lung function
  - decreased blood pressure
  - weight control.

Coronary Artery Disease

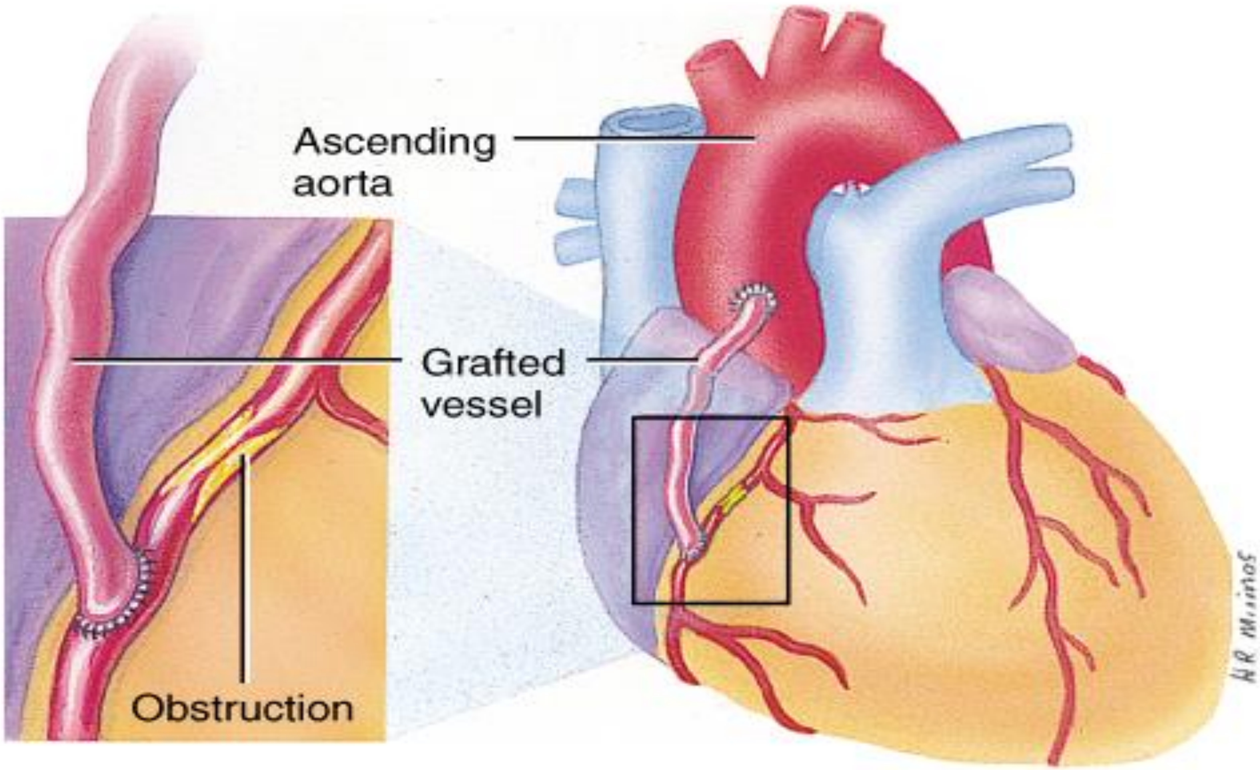


- Heart muscle receiving insufficient blood supply
  - narrowing of vessels---atherosclerosis, artery spasm or clot
  - atherosclerosis--smooth muscle & fatty deposits in walls of arteries
- Treatment
  - drugs, bypass graft, angioplasty, stent

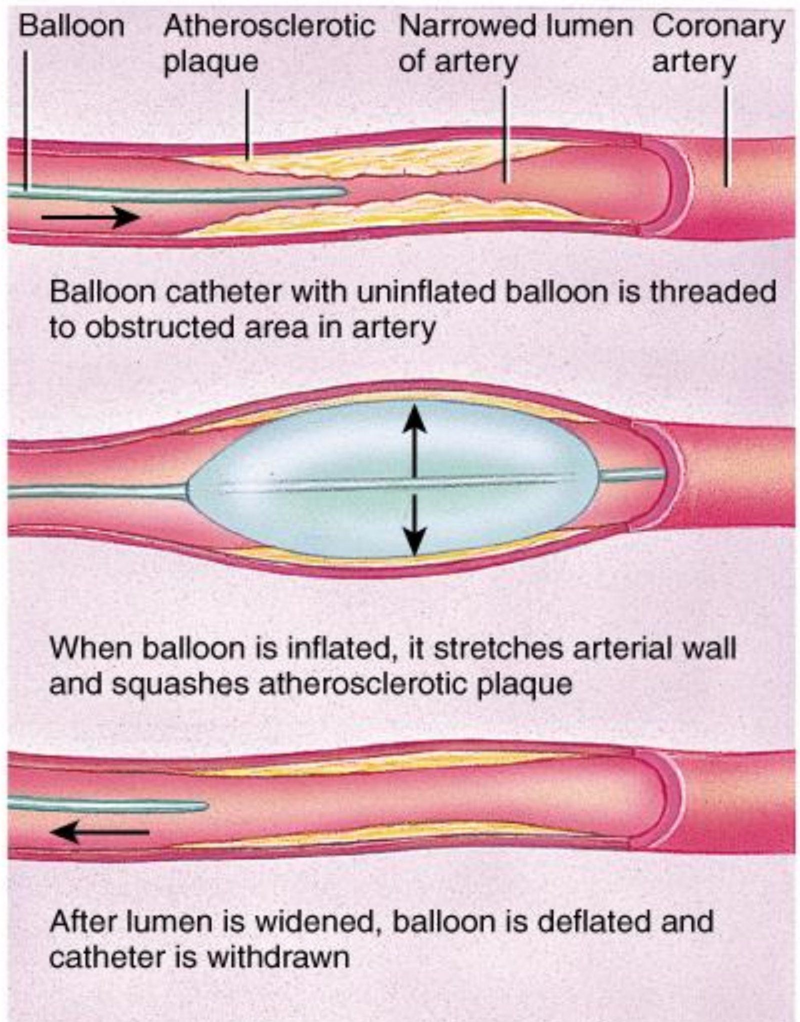
Clinical Problems

- MI = myocardial infarction
  - death of area of heart muscle from lack of O_2
  - replaced with scar tissue
  - results depend on size & location of damage
- Blood clot
  - use clot dissolving drugs streptokinase or t-PA & heparin
  - balloon angioplasty
- Angina pectoris----heart pain from ischemia of cardiac muscle

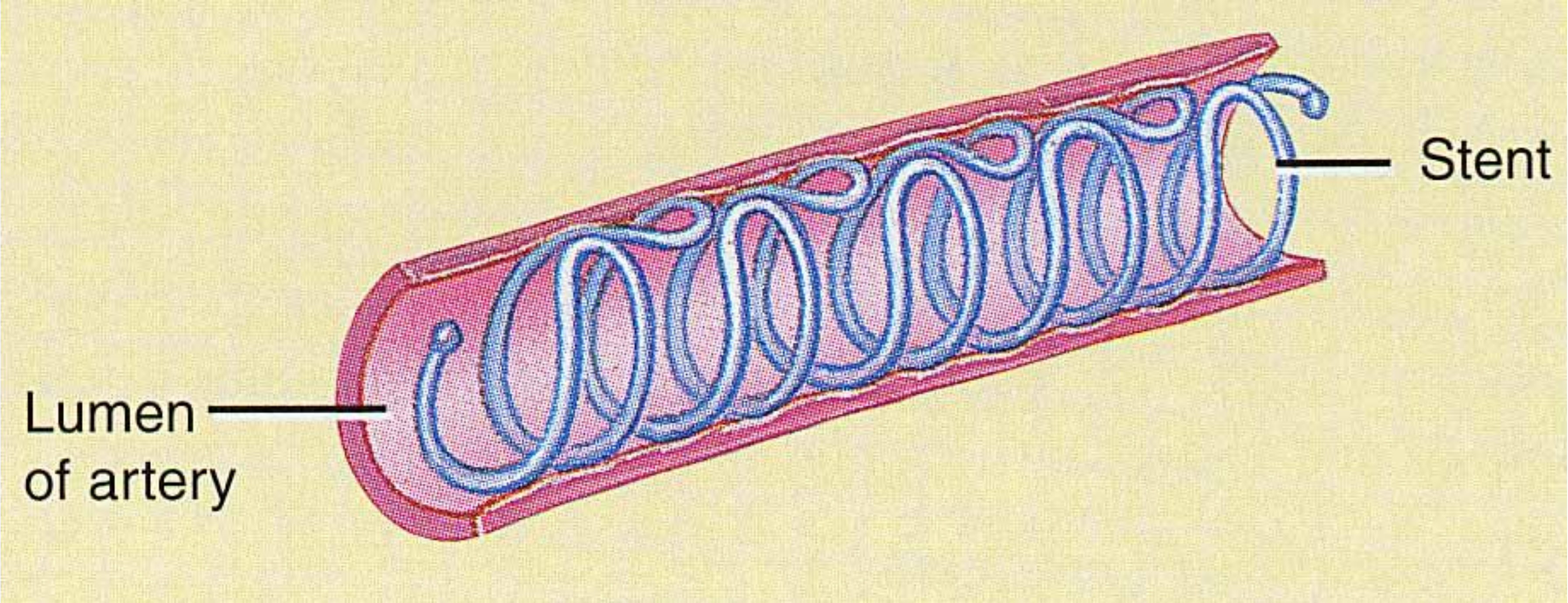
By-pass Graft



Percutaneous Transluminal Coronary Angioplasty



Stent in an Artery



- Maintains patency of blood vessel